Exposure and Recovery with Persistent Randomness Reservations

Ziang Li[1*]

[1]Department of Engineering, Durham University, Durham, United Kingdom.

Corresponding author(s). E-mail(s): 2403871950@qq.com;

Abstract

Persistent implementations can reserve randomness before a cryptographic operation executes. Recovery after compromise therefore depends on the generation and acquisition history of that randomness across restart. We formalize this history through immutable reservation origins and a record interface for acquisition, copying, rewrapping, and later key disclosure. Alias normalization preserves each retained representation's original operation. For fixed public key derivations, a guarded-path theorem exactly characterizes record exposure and its conjunctive acquisition dependencies. We construct a simulator for private images and intents without selected reservation coins, and compose immutable authentication reports in reverse causal order when their transcripts contain earlier report keys. Candidate-local accounting connects these interfaces through an origin-based stopping certificate valid before the final compromise cut is selected. Starting from a uniform recovered contribution, a joint theorem in the compression quantum random-oracle model separates a selected first-message key while retaining confirmation-dependent traffic derivation, real control and acknowledgement verification, and successor-root and next-seed disclosure. We instantiate the origin-based recovery condition in a hybrid post-quantum messaging protocol and establish multi-epoch symbolic message secrecy with inherited roots and asynchronous confirmation. Native executions separate post-restart computation from fresh entropy and show how aggregate rewriting preserves an earlier reservation. Finite-model checks and executable reference corpora validate the exposure semantics. These results identify the original sources, persistent observations, and joint cryptographic continuations required by a recovery argument.

Keywords: Post-compromise security, Persistent state, Randomness reservations, Ratcheting protocols, Record exposure, Provable security, Post-quantum cryptography

1 Introduction

Post-compromise recovery depends on the secrecy of contributions incorporated into later keys. Ratcheting protocols make this dependence precise through the evolution of key-exchange and messaging state [7, 10]. In a persistent implementation, a contribution also has a storage history: its randomness is sampled, a replay record is saved, the cryptographic operation is executed, and its result is published and confirmed. These events can be separated by a crash or a delayed message. A security experiment must account for the secret source across this entire history.

The reservation boundary.

Consider an endpoint that durably reserves the random tape of a KEM operation so that an interrupted call can resume with its original inputs. An adversary acquires this reservation, its wrapping key, and the predecessor root before the key-generation call executes. After a crash, the endpoint resumes the call and completes an honestly authenticated exchange. The captured tape determines the same private key; the public response then determines the KEM contribution and, with the predecessor root, the descendant traffic keys. The resumed computation is a later event in the lifetime of the original secret source.

Future-key exposure already has a precise place in messaging security. SCKA's vulnerable message sets include retained future-epoch keys [1], and Triple Ratchet accounts for next-epoch material sampled while its transmission is in progress [10]. The question here concerns the persistent objects from which such material can be recovered: *which facts about an original operation and the acquisition of its records must a recovery experiment preserve through restart, copying, and later key disclosure?*

There are two related difficulties. First, one reservation may occur in a pending target, a current journal image, and several retained copies. A later wrapping-key disclosure can make an earlier capture readable, even after the honest endpoint has deleted its local copy. Second, a cryptographic reduction may possess a selected operation's public outputs without possessing its random tape. It must still account for public record commitments, complete-record equality, exact retries, reached persistence states, and authenticated reports whose inputs depend on retained private values. These are semantic and simulation obligations of the recovery argument.

Original sources and durable executions.

We associate each reservation with its original owner, operation, session, epoch, and sampling event. We separately identify an immutable session-record version and a physical image carrying that version. Copying or resealing an image preserves the referenced source and record version. Acquisition records the objects obtained by the adversary; readability is evaluated against the complete acquisition and key-disclosure history. This representation makes the secrecy premise stable under changes to physical storage representation while allowing adversarial knowledge to grow.

We study this representation in Q-Periapt Continuity, a hybrid post-quantum messaging construction with persistent rekey operations and asynchronous confirmation. The construction provides a concrete setting in which the reservation, execution, and

confirmation events have distinct operational boundaries. Its journal can also aggregate several sessions, making the distinction between a new image and an existing source directly observable.

Contributions.

The paper establishes three connected results.

1. **A provenance-preserving model of durable exposure.** We define the original-reservation and record-acquisition interface and prove alias normalization with equality of public views and original-record ledgers (Lemma 4.3). The result preserves actual rendering operations and counts distinct plaintext openings by original record, while retaining the separate costs of fresh seals and public cryptographic calls. For fixed public key derivations, an exact guarded-path characterization equates readability with iterative decryption (Theorem 4.4). Its conjunctive guards represent the joint effect of captured ciphertexts and disclosed keys; alternative copies supply additional routes to the same source.
2. **Constructive composition for persistent observations and key derivation.** We construct a table-based image/intent simulator that operates without the selected reservation's bytes after the stated private-commitment replacement (Theorem 5.2). The algorithm preserves original invocation identities, complete intent prefixes, exact replay, and reached durable state, with an explicit nonce-collision loss for byte equality. A reverse-order hybrid theorem composes immutable authenticated reports whose transcripts may contain earlier private report keys (Theorem 5.4). Its reduction uses the causal order to execute later dependent reports without the selected unknown key. Allowed key disclosures remain in the real experiment and contribute their actual scored residual (Corollary 5.5). These interface results connect to the native application through candidate-local accounting (Proposition 5.8). Its score decomposition charges the selected candidate's actual disclosure residual once in the original game, while retaining each candidate's own comparison world. An origin-based stopping certificate remains valid before the final cut and challenge are selected (Lemma 5.6). Starting from a uniform recovered contribution, a joint key-schedule theorem separates the selected first-message key in the compression quantum random-oracle model (Theorem 5.10 and Corollary 5.11). The reduction retains traffic information chosen from confirmation outputs, real confirmation and ACK verification, and disclosed successor roots and next chain seeds. Using exact private-row normalization, the composition makes the next hidden-input reduction possible without leaving an unmodeled parent-row use of its secret (Lemma 5.9).
3. **A multi-epoch recovery condition with separating executions.** We prove symbolic message secrecy when both target reservations follow the compromise cut, the actual target authentication keys and target entropy remain undisclosed, and no acquired target-session version was saved after the cut (Theorem 6.1). The model admits inherited roots, shared key identities, alternating proposers, asynchronous confirmation, and unbounded epochs. The theorem and its reusable dependencies were checked together in Tamarin. Native process-kill executions

at both reservation boundaries reconstruct traffic after honest completion and restart. A delayed-receipt execution exposes epoch-two traffic when the peers' confirmed counters are zero and one; a sibling-session write retains the target's original reservation through a new aggregate image. These executions identify the concrete distinctions captured by the recovery condition.

Supporting validation.

An executable finite-model supplement checks the guarded characterization and its treatment of acquisition, deletion, and late disclosure against independent expected results. A source correspondence maps the image, intent, retry, report, and acquisition operations to the declared interfaces. Separately, pinned SPQR and whole-KEM reference corpora measure reconstructible message sets across communication schedules. Their purpose is to explain how retained material and communication opportunities affect observed exposure. The measurements retain the originating contribution, direction, epoch, and message index, together with each construction's authentication and storage interface.

Security model and organization.

Crash recovery in this paper resumes retained, bound operations; authentication, acquisition, and disclosure are specified by the model in Section 4. The universal recovery result is symbolic. The quantitative results cover record and report simulation, native score accounting, and a joint first-message continuation for the stated finite interfaces. Their cryptographic and disclosure terms remain explicit in the common message experiment. Section 3 describes the construction; Sections 4 and 5 develop the record model and computational connections. Section 6 gives the mechanized recovery theorem and separating executions, and Section 7 validates the exposure semantics. Section 8 connects the recovered-contribution entry to the accepted-operation adapter and the matched protocol comparison.

2 Related work

Ratcheting and fine-grained exposure.

Cohn-Gordon et al. model Signal's key-exchange core as a branching multi-stage authenticated key exchange and prove security for that abstraction [7]. Auerbach et al. develop SCKA with vulnerable epoch sets and derive secure messaging relative to direction- and index-sensitive vulnerable message sets [1]. Their Definition 3.2 includes current chains, already produced future-epoch keys, and pending out-of-order messages; Theorem 3.3 gives the corresponding messaging composition. The object studied here is the history of persistent records from which a contribution can be recovered. Original-reservation provenance connects that history to exposure of the underlying cryptographic operation.

Triple Ratchet combines classical and post-quantum key agreement while distributing larger post-quantum messages across transmissions [10]. Its Remark 4.3 accounts for exposure of next-epoch material already sampled during distribution. This is the relevant comparison for our delayed-receipt execution. Our additional distinction is

between the original source, its durable representations, and the later computation that uses it. Each representation may have its own acquisition time and wrapping keys while referring to the same reserved operation.

Persistence, exact replay, and crash recovery.

The Signal specification discusses deferring ratchet-key generation to shorten the lifetime of sensitive state [16, Section 8.5]. A persistent reservation can fix that state before the deferred call runs. The original-reservation event therefore identifies the lifetime relevant to our freshness condition. Exact replay and a fresh sealing nonce preserve that source identity even when they change the operational or physical representation.

Crash consistency addresses the state recovered after an interrupted update. For example, GoJournal provides verified atomicity for storage applications through concurrent crash specifications [5]. This storage property explains which retained operation an endpoint resumes. Our exposure interface additionally records which copies an adversary acquires and which key disclosures enable their contents. The native reservation executions illustrate the interaction: an operation resumes with its original inputs, and those inputs can also determine the adversary's later derivation. The guarded normalization theorem expresses that relation for the stated encrypted-record fragment.

Session handling and state loss.

Cremers, Jacomme, and Naska analyze how the session-handling layer determines conversation security and give a Tamarin analysis of Signal's Sesame [8]. Cremers, Medinger, and Naska prove incompatibilities between full end-user PCS and specified state-loss resilience requirements [9]. These results make the surviving state and authentication resources essential parts of a recovery claim. Our failure model retains the bound journal and resumes its original operations. Its disclosure conditions follow actual key identities, including a key shared across sessions, and acquisition of the target projection inside an aggregate record. This specifies the persistence and authentication setting of the symbolic recovery theorem.

Admissible network behavior.

Fischlin, Günther, and Janson formalize robustness for channels over unreliable networks and analyze the record layers of QUIC and DTLS 1.3 [12]. Their correctness and security notions account for supported delivery histories. Our model likewise retains delayed receipts and asymmetric progress as ordinary executions. The resulting provenance question concerns the secret inputs retained along such a history. In particular, the delayed-receipt witness is evaluated using both endpoints' state and the captured reservation, preserving their separate confirmed-epoch counters.

Mechanized ratchet analyses.

Cheval, Jacomme, and Richards give an automated Double Ratchet analysis covering out-of-order messages and key-compromise scenarios, including PCS results and forward-secrecy attacks and fixes [6]. Hérouard et al. mechanize computational PCS for a two-party asymmetric ratchet with unbounded steps and a fixed non-adaptive

compromise scenario [13]. Their proof distinguishes non-deduction of intermediate roots from real-or-random security of derived keys. Our symbolic recovery theorem and quantitative interface theorems retain their respective conclusion types. The private-record simulator addresses the separate task of executing a specified storage interface when selected operation bytes are unavailable to a reduction.

Bhargavan et al. study SPQR and its implementation [4]. The authors' public account describes ProVerif protocol models and hax/F* verification of panic-freedom and selected correctness properties of the Rust core [3]. These developments motivate comparison at the level of explicit state and observation interfaces. Our source correspondence records the native constructors and transitions represented by the original-record model. The SPQR reference corpus supplies independently identified executable measurements of retained-secret exposure and communication; its observations keep their source revision and predictor interface.

Key extraction and dependent uses.

HKDF-Extract places its input key material in HMAC's message argument [15]. In the root update studied here, this input remains the secret contribution when the predecessor root is disclosed. Backendal et al. analyze HMAC's swapped PRF use and characterize its feasible input domains [2]. This is the relevant primitive-level distinction for the root-extraction step. Our computational accounting places that use alongside the record, report, confirmation, and traffic interfaces of the same recovery experiment.

3 Construction and the motivating execution

Q-Periapt Continuity combines ML-KEM-768 and X25519 in its ContextBound hybrid KEM and authenticates protocol controls with an ML-DSA-65/P-256 identity. Our analysis follows the persistent session and rekey machinery: an original operation first reserves its entropy, places the resulting token in a retained plan, and later computes or replays the corresponding KEM operation. An exact retry uses the same reservation. The protocol alternates the proposing endpoint across epochs and carries separate send, receive, and confirmed-epoch state.

Fixed cryptographic context.

The combiner hashes a length-delimited encoding of the two component secrets, both public keys and ciphertexts, and domain, suite, policy and application-context fields. On the rekey path, the application context is a domain-separated digest of the complete retained offer wire and is nested with the admitted policy digest. These fields form fixed inputs of the original operation; exact replay retains that binding and its reserved tape. The source correspondence in Appendix A.7 identifies the encoding and reservation constructors used here.

3.1 Reservations and persistent representations

A generation reservation determines the future hybrid key pair. An encapsulation reservation, together with the authenticated offer, determines the corresponding ciphertext

Event	Effect on the target operation	Provenance fact
Reserve	Draw KEM entropy and seal its replay token	Creates an original source
Persist target	Retain the original pending-write bytes	Creates an acquisition opportunity
Apply or retry	Install or replay the retained target	Preserves the source
Sibling write	Re-encode the aggregate journal with a new outer nonce	Preserves unchanged target sources
Confirm	Advance the relevant endpoint's protocol state	Records protocol progress

Table 1 The events tracked by the analysis. Several events may occur in one symbolic transition; the native execution exposes their separate boundaries.

and contribution. The implementation seals these reservations under the host wrapping key. It persists the selected target in a pending write before installing that target as the next journal image. Applying or reopening that write preserves its original contents.

A journal image aggregates state from multiple sessions. Updating a sibling session advances the global image revision and samples a new outer encryption nonce, while the target session's retained reservation can remain unchanged. Consequently, the journal revision, outer nonce, local operation identifier, and original entropy reservation represent distinct identities. We preserve these distinctions in the model and relate them to the native execution boundaries.

3.2 Root and traffic dependencies

Write R_{e-1} for the preceding root, S_e for the hybrid contribution, C_e for the public response core, and O_e, P_e for the complete signed offer and response. With the construction's canonical encodings and domain-separated transcript hashes, its dependency equations are

$$R_e = \mathrm{HKDF}_{\mathrm{SHA256}}\big(R_{e-1}, S_e, d_R \| H(d_C, C_e), 32\big), \tag{1}$$

$$T_e = \mathrm{HKDF}_{\mathrm{SHA256}}\big(sid, R_e, d_T \| ctx \| e \| H(d_{\mathrm{tr}}, O_e \| P_e), 128\big). \tag{2}$$

Here HKDF arguments denote salt, input key material, information, and output length in bytes. T_e supplies two directional traffic chains and two acknowledgement keys. The actual domain strings and encodings are pinned by the source locations in Appendix A.

These equations describe a concrete derivation path. A captured generation reservation and matching wrapping key reconstruct the private key; a subsequent public ciphertext then yields S_e. A captured encapsulation reservation yields the same contribution using its retained offer. Combining S_e with a disclosed R_{e-1} and the public control wires reproduces R_e and the traffic material. Honest admission checks can still reject an operation inside the SDK while an adversary holding these inputs performs the underlying computation independently.

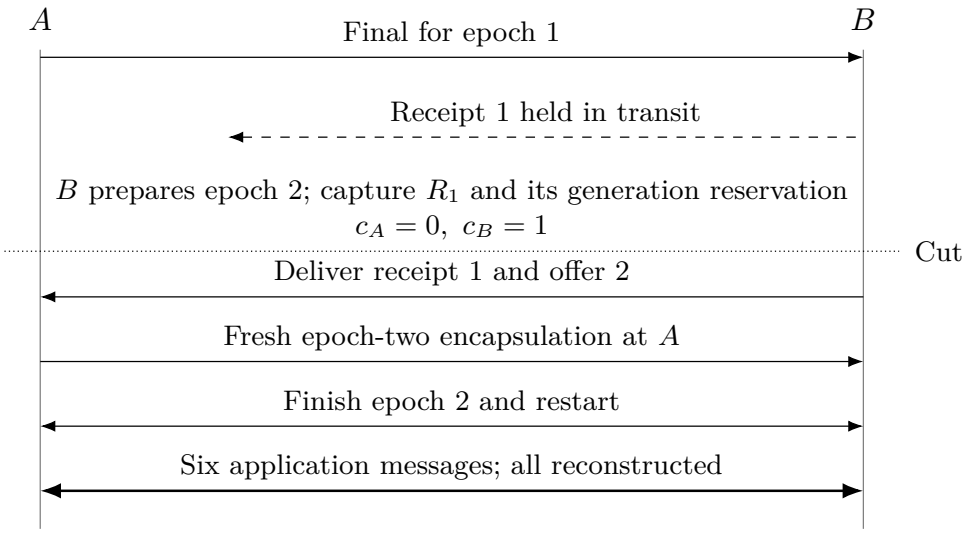


Fig. 1 A retained native disclosure execution. The smaller confirmed counter is zero at capture, while the captured reservation enables epoch-two traffic recovery after the public exchange. The dotted Cut orders the exposure and the subsequent fresh encapsulation; it does not refresh the earlier generation reservation.

3.3 Asynchronous confirmation and a later exposed epoch

Consider the native execution in Figure 1. Endpoint B commits the first receipt, advances locally, and prepares an epoch-two offer before A receives that receipt. At the capture point their local observations are

$$A : (c, s, r, p) = (0, 1, 0, 1), \qquad B : (c, s, r, p) = (1, 1, 1, 2),$$

where c, s, r are confirmed, send, and receive epochs, and p is the pending target epoch. The snapshot of B contains R_1 and the original generation reservation for epoch two. After delivery resumes, those captured inputs and the public exchange reconstruct epoch-two traffic.

The resulting counterexample targets the proposed bound $\min(c_A, c_B) + 1$: its value is one, yet epoch-two traffic is exposed. The protocol's asymmetric progress is legitimate. Recovery reasoning must therefore follow the retained inputs and their descendants rather than identify the two endpoints' confirmation counters.

4 Reservation origins, persistent records, and recovery

An interrupted cryptographic operation can retain its random tape before it has produced a key or a ciphertext. Its later execution and confirmation therefore have to be related to the original reservation. We make this relationship explicit at two levels: the origin of a cryptographic operation, and the origin of a saved session-state version. A third identifier describes the physical image that carries that version. This

Table 2 Principal objects in the record model. The three identities separate an original random draw, an immutable session-state version and its physical storage representation.

Notation	Meaning
o, ρ_o	Original invocation and its reserved random tape
v, X_v	Immutable session-state version and its payload
h	Physical image or pending object carrying one or more versions
κ	Identity of actual key material, shared by equal keys
$\mathsf{res}(o)$	Event at which the original random tape is drawn
$\mathsf{save}(v)$	First retention of v on the durable acquisition surface
$\mathcal{C}_t, \mathcal{L}_t$	Acquired and readable original versions at prefix t

distinction lets us account for a target record copied by another session's write, an unapplied pending image, and a ciphertext acquired before its wrapping key.

The model is an interface model for an asynchronous protocol machine $\mathcal{P}$. The adversary schedules public messages, process loss, honest recovery, and the disclosure operations specified below. Application deliveries are private to the honest applications. Capture grants read access to the stated assets; it does not grant an additional operation for replacing the authoritative journal. An implementation with adversarial rollback or forking requires those transitions in its storage machine. The protocol, its cryptographic primitives, and its public outputs are parameters of the record-normalization lemma. Fix finite execution, record-size, and query budgets; the wrapper's bookkeeping is polynomial in their total representation size.

Table 2 summarizes the principal objects.

4.1 Three origins and their event order

Definition 4.1 (Reservation origin) A reservation origin is a fresh invocation index o, together with

$$\big(\mathsf{kind}_o, \mathsf{owner}_o, s_o, e_o, B_o, \mathsf{res}(o)\big).$$

Here kind_o distinguishes generation, encapsulation, and any other reserved operation; s_o, e_o identify its session and epoch; B_o is the full canonical binding, including the admitted policy and operation inputs; and $\mathsf{res}(o)$ is the event at which its random tape ρ_o is drawn. Execution or exact replay of o consumes the same ρ_o and the same binding. A new draw receives a new origin even if a failed attempt's external operation identifier is reused.

The ideal source in this model draws independent tapes of the prescribed lengths. Explicit entropy-disclosure queries reveal designated tapes. Independence is a distributional assumption about this source, rather than a consequence of a later API call. The generation and encapsulation budgets G, C count original draws, including draws in attempts that subsequently fail. Exact retries consume no new slot. Independently sampled operations retain distinct origins even if some public outputs happen to coincide.

A record version v has an immutable payload X_v and semantic metadata

$$\big(\mathsf{subject}(v), \mathsf{role}(v), \mathsf{epoch}(v), \mathsf{phase}(v)\big).$$

The subject identifies the session whose state is represented, rather than the session that most recently wrote the enclosing journal. A version can refer to several reservation origins and to roots or traffic state derived from earlier origins. Preparing a version allocates its handle. The event $\mathsf{Saved}(v)$ occurs when that version first enters the durable acquisition surface, and its time is denoted $\mathsf{save}(v)$. In particular, an intact target in a durable pending-write intent is saved before the corresponding image is applied. A storage operation may have made this transition even when its caller receives an unknown-outcome error.

A physical object h has a separate creation or sealing index. It may carry one version, several sessions' versions, an exact earlier image, or a pending target. An unchanged version keeps its v and $\mathsf{save}(v)$ when it is copied, archived, applied from a pending intent, or included in a newly sealed aggregate. A new semantic session-state version receives a new v even when some of its fields are unchanged. These rules distinguish the original reservation from each session-state update that retains it.

The following transition rules specify the provenance discipline.

1. *Reserve and execute.* Reserve creates o at the actual draw. Execution uses its retained binding and tape. Unknown-outcome reconciliation keeps that origin; a definite new attempt allocates another.
2. *Prepare and retain.* Preparing v fixes X_v and its provenance. Retaining the first intact image or pending target containing v emits $\mathsf{Saved}(v)$. An attempt that retains no such object has no successful acquisition event.
3. *Copy, apply, and reseal.* An exact copy preserves the complete object and its sealing origin. Resealing creates a new sealing invocation and its actual new wrapper randomness, while preserving the referenced record versions. Applying an existing pending target copies its retained bytes.
4. *Crash and resume.* Process loss removes volatile references. Recovery reconstructs the retained handles and their original provenance. It does not turn a retained reservation into a new draw.
5. *Confirm and delete.* Confirmation changes protocol state and may authorize particular host-side deletions. Neither action changes a reservation's origin or removes an adversary's previously acquired copy.

For an original reservation actually retained in v, causality requires

$$\mathsf{res}(o) \;\le\; \mathsf{save}(v). \tag{3}$$

A later physical object's creation time gives an additional timestamp, not a replacement for either side of (3). The interval between drawing ρ_o and first retaining it remains explicit. The symbolic reservation-and-save transition can represent that interval only through a separately justified source correspondence.

4.2 Acquisition, delayed readability, and key identity

Wrapping-key handles resolve to identities of actual key material. Two session labels that designate the same key resolve to the same identity κ. Disclosure through either label therefore enables every acquired wrapper protected by κ. A rotation creates a different identity when it creates different key material. Signing-key identities are treated in the same way across sessions and roles.

For each intact physical object h, its construction records a finite set

$$\mathsf{paths}(h) = \{(v, \gamma)\},$$

where γ is the set of wrapping-key identities on a recoverable serialization path to X_v. Multiple paths to one version are permitted. An unprotected record has $\gamma = \varnothing$. Nesting a wrapper adds its key identity; aggregation unions its members' paths. An exact ciphertext copy keeps its paths. Resealing decoded records reconstructs the paths under the new wrapper and retains the same v values. Thus resealing decoded plaintext and wrapping an old ciphertext are distinct operations.

Only intact, recoverably embedded record payloads create these paths. A public hash or MAC of a private record is a public computation of that record, not a path to the record itself. The renderer introduced below preserves that computation and its output. This distinction prevents a public head query from being relabeled as a snapshot merely because its computation used private bytes.

Definition 4.2 (Record acquisition and readability) A successful $\mathsf{Capture}(h)$ returns the actual retained bytes of h and adds an immutable copy to the adversary's archive $\mathcal{A}_t$. Let $\mathcal{K}_t$ be the wrapping-key identities disclosed by time t. Define

$$\mathcal{C}_t = \{v : \exists h \in \mathcal{A}_t,\ \exists \gamma,\ (v, \gamma) \in \mathsf{paths}(h)\}, \tag{4}$$

$$\mathcal{L}_t = \{v : \exists h \in \mathcal{A}_t,\ \exists \gamma,\ (v, \gamma) \in \mathsf{paths}(h)\ \wedge\ \gamma \subseteq \mathcal{K}_t\}. \tag{5}$$

Each capture records $\mathsf{AcquireOriginal}(v, t)$ for every version in its support, including versions already acquired through another path. An unsuccessful request for unavailable state returns its ordinary error and adds no acquisition.

Both sets describe the specified disclosure interfaces, using provenance and key identities. They do not test whether the adversary has guessed a secret. The read set can grow without another capture: if h was obtained at t_1 and its final missing wrapping key is disclosed at $t_2 > t_1$, the corresponding versions enter $\mathcal{L}_{t_2}$. Host deletion at an intermediate time leaves h in $\mathcal{A}_{t_2}$. Acquisition, by contrast, already occurred at t_1.

An additional key that is explicitly recoverable from an opened record can be included by a declared extraction rule. In this extension, iterate (5) and the declared key-extraction rules to their least fixed point after each event. Each rule identifies its input bytes, its deterministic extraction algorithm, and the resulting actual key identity. Only rules with available inputs fire. This accounts for nested protection without assuming that inaccessible keys have already been learned.

Signer-file capture is a separately typed disclosure. For correspondence with the retained symbolic theorem, it records exposure of the actual signing key conservatively

at acquisition, as does that theorem's signer-file rule. The readability ledger can additionally record when the file's contents become decryptable. In particular, a wrapping-key disclosure cannot be scoped to traffic state when the declared capture also includes the matching signer file.

Constructive exposure.

From public protocol inputs, explicit leaks, and readable record payloads, we form a labelled replay-dependency graph. Its edges are specified algorithms: unseal a reservation with its matching key; regenerate a key from its original tape; replay encapsulation on the bound offer; decapsulate a published ciphertext; derive a root; and derive the direction-specific traffic and ACK material. Directly stored roots, skipped keys, and plaintext enter through their own record fields. For the implemented root schedule, an edge is

$$(R_{e-1}, S_e, C_e) \longmapsto R_e = \mathsf{HKDF}\big(R_{e-1}, S_e, D_{\mathrm{root}} \| H_{\mathrm{core}}(C_e), 32\big).$$

Closing these explicit edges over the nodes of the bounded execution yields a computable set of message keys with replay witnesses. It supplies positive exposure certificates; the confidentiality game below is the separate criterion for messages outside the declared disclosure conditions. Confirmation counters belong to the public progress state and do not reset this graph.

4.3 An original-record implementation of the copy interface

We now isolate a property of record management that is independent of KEM secrecy. The following two storage machines use the same protocol machine $\mathcal{P}$, random-tape source, scheduler, and cryptographic interfaces.

The *literal-copy machine* materializes the immutable payload X_v in each indicated serialization position. Its objects include current images, pending targets, and retained historical copies. It records the origin tags alongside these values as private bookkeeping.

The *original-record machine* keeps one immutable table entry $\mathcal{O}[v] = X_v$ and represents each serialization position by a reference to v. Its wrapper maintains the physical-object directory, pending states, adversary archive, origin tags, and key identities. It has the following specified interfaces.

1. $\mathsf{RegisterOriginal}(v, X_v)$ is a private write by the honest machine. It fixes one value at a fresh v. The wrapper receives the handle and the provenance metadata.
2. $\mathsf{RenderOriginal}(\theta)$ evaluates the supplied serialization recipe θ by dereferencing $\mathcal{O}$. A recipe specifies literal public fields, record references, references to already rendered objects, and each sealing, hash, MAC, or other admitted byte-processing call. A fresh sealing call uses a fresh invocation index and the same prescribed coin distribution as the literal-copy machine. An exact-object read or replay returns its cached original bytes.
3. $\mathsf{AcquireOriginal}(v, t)$ records acquisition of that version. This is an event, not an assertion of decryption. A capture's externally returned bytes come from the rendered object cache.

4. RevealOriginal(v) returns the original payload X_v. The wrapper uses this interface for a plaintext record-opening response only after a captured path is enabled by the disclosed keys, and caches the answer for subsequent openings.

The renderer is an explicit trusted interface in this intermediate experiment. It has access to the original payload table and evaluates the stated byte operations, including operations on undisclosed payloads. Consequently, it can produce real ciphertexts, commitment digests, and report MACs. This definition does not supply an algorithm for producing those bytes from an unknown KEM-challenge tape. It gives a precise location for the subsequent cryptographic replacement of that interface.

The operational contract has three further requirements. First, all publicly visible payload-dependent bytes and statuses are obtained through the specified renderer or through $\mathcal{P}$'s common protocol computation. Memory addresses and host timing are not additional outputs of this interface. Second, actual byte-operation order, error responses, partial transaction outcomes, and cryptographic-oracle calls are retained. Caching an exact object does not discard a native hash-validation call, observable retry, or write attempt: such an action remains a renderer or protocol action. Third, the reference metadata and the key-identity relation are available to the wrapper without inspecting a selected hidden reservation's coins. These are checkable interface conditions; none asserts the secrecy of a record, a root, or a message.

Lemma 4.3 (Alias normalization) *For the operations and renderer contract above, every adversary $\mathcal{A}$ interacting with the literal-copy machine has an online wrapper $\mathcal{B}$ interacting with the original-record machine such that their public views have the same distribution. Their reservation-origin ledgers, record-save events, acquisition events, and read sets are identical after projection onto the original handles.*

The wrapper makes at most $|\mathcal{L}_T|$ distinct RevealOriginal *calls by terminal time T. Repeated physical copies, captures, and openings do not multiply this number. The wrapper preserves every original reservation and every fresh sealing invocation, so it does not merge their respective cryptographic resource counts.*

The statement also holds for an adversary with quantum private state and quantum access to common public cryptographic oracles, provided the storage interface is classical and both experiments retain the same oracle calls.

Proof Maintain a map from every literal record occurrence to its immutable handle v, and from every native object identifier to a rendered-object entry. Couple the protocol random tapes, sealing coins, scheduler choices, and public cryptographic oracles between the experiments. The invariant is that dereferencing every original-record recipe produces the corresponding literal-copy bytes, with the same semantic provenance and physical disposition.

For a new record, register its payload once and extend the occurrence map. For a fresh rendering, submit the same recipe and byte-operation sequence to RenderOriginal. Dereferencing gives the same inputs to every sealing, digest, MAC, validation, and output operation. Coupled coins and common oracles therefore give the same bytes, statuses, and state changes. This step retains a write that became durable before returning an error; the caller's status does not overwrite the persistence state.

An exact copy or pending-target application copies the object reference and its original cache entry. Aggregation collects the corresponding original references, and resealing evaluates

the new recipe using those references. Each operation preserves the session-record handles and their save times. It follows by induction on the recipe constructors that $\mathsf{paths}(h)$ is the same in both machines. A public hash or MAC constructor contributes its common output, rather than a spurious record-acquisition path.

On successful capture, output the cached object bytes in both experiments and append its immutable descriptor to the adversary archive. For each version in its support, emit the same acquisition event at the actual capture time. Preserve the ordinary error for an unavailable object. When a matching wrapping key is disclosed, use the common key identity and reevaluate the enabled paths in the complete adversary archive. For an opening response, obtain X_v from $\mathsf{RevealOriginal}$ on its first use and from the wrapper's cache thereafter. Every such call has $v \in \mathcal{L}_T$. Shared-key aliases and later key acquisition are thus handled without another copy of X_v. The optional key-extraction extension preserves the invariant by induction on the least-fixed-point rounds.

Host deletion removes only host references. The adversary archive and already returned payloads persist. Crash and resume use the same retained object identifiers; neither operation recreates a reservation or changes a save event. Ordinary protocol actions are performed by the same $\mathcal{P}$ with the same results. This completes the induction, including adaptive choices based on any preceding response.

For quantum private state, identical classical responses and the same public oracle transformations induce the same transformation of that state at each interaction. The argument introduces no measurement of a quantum oracle query and no oracle-dependent advice. Finally, caching each plaintext opening by v proves the query bound; reservation and fresh-seal indices were preserved separately throughout. □

The lemma places multiplicity in the correct ledger. An adversary may retain many ciphertext aliases of one version, and those ciphertexts may have many different public heads. All such byte-level observations remain in the view. The repeated payload uses one original-record opening, whereas each genuinely fresh seal and each public hash or MAC computation keeps its own resource accounting.

4.4 Explicit guarded normalization of record readability

The path and readability definitions above already account for copying and later key disclosure. We now give their explicit guarded normalization and an executable realization for a record-decryption specialization. Here key derivation has fixed public edges independent of recovered plaintext. The optional plaintext-dependent key-extraction extension remains as defined above and is outside this specialization. The set Ω_{src} below contains reservation-source identities; it is distinct from saved record versions and from the original-payload table $\mathcal{O}[v]$. Likewise, $\mathcal{E}_t^{\mathrm{src}}$ projects derivability onto protected source values, whereas $\mathcal{C}_t$ and $\mathcal{L}_t$ track acquired and readable versions. The graph is a fixed representation of admitted retained objects, not a replacement for the provenance transition rules or the byte renderer.

A source identifier names one original sampling operation. An acquisition history records what the adversary obtains, including objects obtained at different times. These are distinct objects of the model: a copy preserves the former, while an additional acquisition can change the latter.

Immutable record graph.

Let $\mathcal{R}$ be a finite set of immutable encrypted record objects. Each $r \in \mathcal{R}$ has a matching decryption-key identity $\kappa(r)$, a set of embedded child records $\mathrm{child}(r)$, and a set $\mathrm{orig}(r)$ of protected source values in its plaintext. Containment is acyclic. Record identities distinguish ciphertext objects; source identities distinguish original sampling operations. Several records can contain the same source. A rewrapped record has a new record identity and may have a new key, while retaining its source identity. Key identities are equivalence classes of equal actual keys, rather than names assigned independently by separate API objects.

Let $\mathcal{D} \subseteq \mathcal{K} \times \mathcal{K}$ be fixed public key derivation edges. An edge (k, k') means that possession of k permits deterministic computation of k'. This interface represents, for example, derivation of an operation-wrapping key from the journal key. In the theorem below, these edges do not depend on plaintext recovered from records. Write D^* for their reflexive transitive closure.

Let $\Omega_{\mathrm{src}} = \bigcup_{r \in \mathcal{R}} \mathrm{orig}(r)$ be the declared source domain. At a trace prefix t, let $A_t \subseteq \mathcal{R}$ be all record objects ever acquired by that prefix, $K_t \subseteq \mathcal{K}$ all directly disclosed keys, and $\mathcal{S}_t^{\mathrm{dir}} \subseteq \Omega_{\mathrm{src}}$ all directly disclosed source values, represented by their source identities. These are cumulative sets. The decryption-key closure is

$$K_t^* = \{k' : \exists k \in K_t,\ (k, k') \in D^*\}.$$

The concrete derivation rules are

$$\frac{k \quad (k, k') \in \mathcal{D}}{k'}, \qquad \frac{r \quad \kappa(r)}{c}\ (c \in \mathrm{child}(r)), \qquad \frac{r \quad \kappa(r)}{o}\ (o \in \mathrm{orig}(r)).$$

Their least closure starts from $A_t, K_t, \mathcal{S}_t^{\mathrm{dir}}$. A fact o in the source sort means that the protected value of source o is derivable; it does not mean that a public identifier has become known. The source projection of this closure is denoted $\mathcal{E}_t^{\mathrm{src}}$.

Guarded routes.

For every path $p = (r_0, \ldots, r_\ell)$ in the containment graph ending at a record with $o \in \mathrm{orig}(r_\ell)$, define

$$g(p) = \{\kappa(r_0), \ldots, \kappa(r_\ell)\}.$$

The normalized table contains a route $(r_0, g(p), o, p)$ for every such path. Its exposure expression is

$$\widehat{\mathcal{E}}_t^{\mathrm{src}} = \mathcal{S}_t^{\mathrm{dir}} \ \cup\ \{o : \exists (r, g, o, p),\ r \in A_t \ \wedge\ g \subseteq K_t^*\}. \tag{GP}$$

Alternative records and paths are disjoined, while all keys on one path are conjoined. Equal key identities occur only once in the guard set.

Theorem 4.4 (Exact guarded normalization) *For every finite record graph and derivation interface above, and every acquisition prefix $(A_t, K_t, \mathcal{S}_t^{\mathrm{dir}})$, the concrete and normalized exposure sets agree: $\mathcal{E}_t^{\mathrm{src}} = \widehat{\mathcal{E}}_t^{\mathrm{src}}$. The equality applies to all prefixes of an adaptively chosen history.*

Proof Key derivations depend only on disclosed keys and the fixed relation $\mathcal{D}$, so their least closure is exactly K_t^*. For the forward inclusion, take a source $o \in \mathcal{E}_t^{\mathrm{src}}$. If it was directly disclosed then $o \in \mathcal{S}_t^{\mathrm{dir}}$. Otherwise its first derivation decrypts a record r_ℓ with the matching key. If that record was not acquired directly, its first availability follows by decrypting a parent. Follow these first derivations backwards. The finite acyclic record graph makes this process terminate at an acquired record r_0. Every intervening key lies in K_t^*, yielding a route satisfying (GP). For the reverse inclusion, take a satisfied route. Starting with its acquired first record, successively decrypt along the path. Every required key is in K_t^*, so each child and finally o is in the concrete least closure. Direct disclosures are in that closure by definition. Both inclusions hold for each fixed prefix; no independence or advance choice of the acquisition schedule was used. □

Efficient evaluation.

Equation (GP) is a declarative normal form: a compressed record DAG can have exponentially many paths. An efficient leakage evaluator instead computes the iterative decryption and key-derivation fixed point, without enumerating those paths. Let N be the total number of record, key and source atoms, and let M count rule and payload incidences. There are at most N strict-growth rounds; repeated rule scans therefore take $O(N(N + M))$ time and $O(N + M)$ space under standard indexed set and dictionary representations, with source-dictionary operations included in this accounting. The finite checker materializes the route table only for its declared small fixtures, to compare that normal form against the forward evaluator.

Corollary 4.5 (Copies, deletion and late disclosure) *Exposure is monotone under additional acquisitions, direct disclosures and key disclosures. Copying or rewrapping a source adds routes to the same source identity. Local deletion of a record previously acquired does not remove its routes from adversarial knowledge. In particular, a later matching-key disclosure can expose a source in a ciphertext acquired before that disclosure and deleted locally in the meantime.*

Proof The cumulative sets and their key closure only grow, and every predicate in (GP) is positive. A source-preserving copy changes a route's record identity or key guards but not its terminal source. Local deletion restricts future acquisition from that local store; it removes no member of A_t and therefore cannot invalidate a satisfied route. □

Interface significance.

The theorem gives an exact finite normalization of the chosen record-decryption semantics. It supports a precise original-operation leakage oracle: a query for source o can be evaluated against the complete acquisition history, with a path and its key guards as a witness. The cryptographic transition from native ciphertexts and public commitments to this interface has a separate simulation obligation. The derivation rules do not model guessing, cryptanalysis, timing leakage or a quantum decryption oracle. Accordingly, absence from $\mathcal{E}_t^{\mathrm{src}}$ is recorded as non-derivability by these rules; message indistinguishability outside an exposure set is supplied by a security theorem for the surrounding protocol.

Native shape and finite checks.

At the fixed research source, operation reservations are encrypted under a wrapping key derived from the journal key, and tokens are retained within encrypted durable images or pending targets. The corresponding route therefore includes the image key and the token key, with their actual derivation edge. A fresh aggregate image can preserve an older token and its original source. Public image-head digests are distinct observations from the record objects needed by the decryption rules. The executable supplement independently evaluates iterative decryption and guarded-route normalization, exhausts all acquisition/key/direct-source subsets for its declared finite graphs, and checks histories with late key disclosure and local deletion. Its counts and source hash are in the accompanying machine-readable result; those checks validate the implemented finite model and supplement the general proof above.

Why exposure must retain conjunctive guards.

The following elementary separation explains the conjunction in (GP).

Proposition 4.6 (Acquisition exposure is not additive) *Let r be a retained-record atom, k a wrapping-key atom, and s a protected-seed atom associated with an origin identifier o. For $A \subseteq \{r, k\}$, let $\mathrm{Cl}(A)$ be the least set containing A and closed under the rule*

$$\{r, k\} \implies s.$$

There are no other rules in this fragment. Define the exposed-origin set by

$$E(A) = \{\, o : s \in \mathrm{Cl}(A) \,\}.$$

There is no fixed function $f : \{r, k\} \longrightarrow \mathcal{P}(\{o\})$ satisfying

$$E(A) = \bigcup_{x \in A} f(x) \qquad \text{for every } A \subseteq \{r, k\}.$$

Proof The closure rule gives

$$E(\{r\}) = E(\{k\}) = \varnothing, \qquad E(\{r, k\}) = \{o\}.$$

Exactness for the two singleton acquisitions forces $f(r) = f(k) = \varnothing$. Their union is then empty for the joint acquisition, contradicting $E(\{r, k\}) = \{o\}$. □

Meaning of exposure.

The proposition concerns derivability in an explicitly specified rule system. The identifier o may itself be public; its membership in $E(A)$ records derivability of its protected seed, not discovery of the identifier. Derivability excludes guessing and is not a claim that a computational adversary has zero success probability. As a separate probabilistic illustration, let S, W be independent uniform λ-bit strings and $C = S \oplus W$, with no additional observation depending on S or W. Then

$$H(S \mid C) = H(S \mid W) = \lambda, \qquad H(S \mid C, W) = 0.$$

From either observation alone, any estimator of S succeeds with probability at most $2^{-\lambda}$; from both, $S = C \oplus W$ recovers it exactly. This illustration does not instantiate the record encryption used by the protocol.

Consequence for provenance routes.

A source label and an exposure decision serve different purposes. Copying a retained record preserves the source label, while subsequent acquisition of a matching key may change exposure. A guarded-route representation retains this distinction: alternative acquired records supply alternative routes to the same source, and every route retains the conjunction of its required keys. Exposure is therefore evaluated from the joint acquisition and key-derivation history. The proposition does not rule out stateful labels, history-dependent vulnerability sets, or conservative per-object overapproximations; it rules out an exact, fixed, per-object union in the presence of the stated conjunctive dependency.

4.5 A recovery experiment with acquisition-based eligibility

Definition 4.7 (Fixed-target recovery experiment) Fix a protocol profile $\mathcal{P}$ and finite execution budgets, together with a designated session s, recovery cut c, target epoch e, and candidate generation and encapsulation reservation handles g_e, a_e. Handles may be named by original invocation slots before those invocations occur; a missing invocation will fail the reservation-event test below. Public protocol messages may be delayed, duplicated, reordered, or replaced. The adversary can capture the declared retained objects, reveal wrapping keys, reveal actual signing keys or signer files, and reveal designated reservation tapes. The cut exposes the selected wrapping keys; preceding roots may already be public or disclosed by the stated prior captures. These disclosures have their exact scope in the event ledger.

For one fresh logical application send, the adversary supplies equal-length messages (m_0, m_1). The challenger samples $b \leftarrow \{0,1\}$ and runs the ordinary sending transition on m_b. Rejection or an unknown outcome keeps its ordinary protocol behavior. A retained accepted send is identified by (s, e, d, j), including direction d and the epoch-scoped logical index j; exact retries reuse its original ciphertext. The challenge is valid only when that logical send is accepted in the designated epoch. Subsequent protocol actions and disclosures continue until the adversary outputs b'.

Write ι_A, ι_B for the two actual signing-key identities configured for the target session. The predicate $\mathsf{ReservedFor}(s, e, g_e, a_e)$ checks the occurrence of the named honest generation and encapsulation reservation events in the target epoch's local durable plans, with their target session and roles. It checks event metadata, without asserting that their outputs form the accepted target exchange. Eligibility uses the following event conditions over the complete execution:

$$\begin{aligned}\mathsf{Eligible}_{s,e,c}(\tau) \iff\ & \mathsf{ValidTarget}(\tau) \ \wedge\ \mathsf{ReservedFor}(s, e, g_e, a_e) \\ & \wedge\ c < \mathsf{res}(g_e) \ \wedge\ c < \mathsf{res}(a_e) \\ & \wedge\ \iota_A, \iota_B \notin \mathsf{IdentityExposed}(\tau) \\ & \wedge\ \mathsf{EntropyExposed}(s, e; \tau) = \varnothing \\ & \wedge\ \nexists v \in \mathcal{C}_T : \mathsf{subject}(v) = s \ \wedge\ c < \mathsf{save}(v). \end{aligned} \tag{6}$$

Here $\mathsf{EntropyExposed}(s, e; \tau)$ records the target epoch's designated entropy disclosures. The separate event Attr states that these named reservation events form the accepted target's unique usable generation/encapsulation pair, with an adapter-computable

association to the original slots. In particular, it connects the generation public key, the encapsulation input, the authenticated response, and the target's subsequent key schedule. This matching is a source/authentication obligation, rather than an additional eligibility filter: a failure of Attr can occur while $\mathsf{Eligible}_{s,e,c}$ holds and is charged in the computational accounting.

The final line of (6) concerns every saved version of session s, across its roles and epochs. It uses acquisition $\mathcal{C}_T$, including protected bytes, rather than delayed readability $\mathcal{L}_T$. Other epochs' entropy disclosures retain their ordinary behavior. Both peers' confirmation is not an additional send prerequisite.

Define the unconditional centered advantage by

$$\mathsf{Adv}^{\mathrm{rec}}_{\mathcal{P}}(\mathcal{A}) = \left|\mathbb{E}\left[\mathbf{1}_{\mathsf{Eligible}_{s,e,c}(\tau)}(-1)^{b\oplus b'}\right]\right|. \tag{7}$$

Invalid or ineligible executions contribute zero. This definition retains failures and adaptive disclosures in the probability space and avoids conditioning a reduction on an uncontrolled eligibility probability. If every execution with $\mathsf{Eligible}_{s,e,c} = 0$ is completed by an independent fair guess, the conventional advantage over one half is

$$\mathsf{Adv}_{\mathrm{msg}} = \left|\Pr[\text{completed guess is correct}] - \tfrac{1}{2}\right| = \tfrac{1}{2}\,\mathsf{Adv}^{\mathrm{rec}}_{\mathcal{P}}(\mathcal{A}).$$

The computational accounting uses this latter convention.

The adaptive version retains one application challenge and permits the candidate selector to choose its coordinates and reservation-event handles from the observed interaction, rather than fixing them at setup. Eligibility requires those coordinates to identify the challenge's accepted logical send; selection never transfers its test bit to another send. Eligibility and the usable-pair attribution are evaluated on that experiment's complete history. An origin-pair reduction fixes a slot pair in $[G]\times[C]$ and scores the event that the final selector attributes the target to that pair. The selector and disclosure predicates are recomputed in each comparison world. No independence between that selection and the transcript is imposed.

The record restriction in (6) is the one used by the retained multi-epoch symbolic theorem: its $\mathsf{Snapshot}$ action outputs a protected record, while Cut separately releases the wrapping keys. The theorem establishes symbolic plaintext secrecy with both original reservations after the cut, key-scoped identity exclusions, the target entropy exclusion, and the stated post-cut saved-record acquisition exclusion [17]. Its joint replay verifies the main lemma and all 24 reusable dependencies. Equation (7) specifies the computational message experiment to which the cryptographic modules are to be connected; the alias lemma establishes its record-interface normalization.

Corollary 4.8 (Safe stopping at a forbidden acquisition) *Consider the fixed-target experiment of Definition 4.7. Suppose an actual successful capture contains a target-session version v with $c < \mathsf{save}(v)$, identified by the original-record metadata. For any terminal bit D, replacing the remainder of that execution by output zero leaves*

$$\Pr[\mathsf{Eligible}_{s,e,c}(\tau) \wedge D = 1]$$

unchanged. Replacing its terminal score by zero also leaves the expectation inside (7) *unchanged.*

In particular, if a captured target-session version retains the selected reservation $o \in \{g_e, a_e\}$, then $c < \mathsf{res}(o)$ and (3) establish this stopping condition. The wrapping key may be obtained before or after the capture.

Proof The successful capture inserts v into $\mathcal{C}_t$ immediately. Its save time and subject are immutable, and acquisition remains recorded after host deletion, resealing, confirmation, or process loss. Thus the last conjunct of (6) is false for every continuation. Each such continuation already contributes zero to both quantities. On executions that never reach this condition, the experiment is unchanged. The final assertion follows from $c < \mathsf{res}(o) \leq \mathsf{save}(v)$. □

The stopping test uses the successful acquisition event and its record metadata; it can run before the response's private bytes are serialized by a challenge adapter. It is not an SDK rejection of the capture. A public digest, an unrelated record, or an unsuccessful capture keeps its real behavior. In particular, difficulty computing a permitted digest or report MAC does not trigger this corollary. For adaptive target selection, stopping is applied to the corresponding fixed-candidate event in the reduction. One possible candidate becoming ineligible does not justify terminating a game that can still select a different eligible target.

Two short executions illustrate why both kinds of origin are necessary. If v was saved before c and a sibling write reseals it after c, the acquisition still refers to the original pre-cut v. The new global image time alone does not falsify the final conjunct of (6). If instead v was saved after c, a later write attributed to a sibling does not change its subject or make its capture eligible. Likewise, a pre-cut generation reservation retained through a delayed receipt remains pre-cut when its KEM computation completes later. Its reservation condition, rather than the slower endpoint's confirmed counter, determines that case.

4.6 Connection to the cryptographic argument

The normalization theorem fixes the original objects whose bytes, acquisition, and key identities the reduction must represent. The next section gives constructive image/intent and report interfaces for a selected KEM operation whose public outputs are available while its original random tape is opaque. Their algorithms preserve original-object caches, public commitments, exact retries, and actual durable outcomes. The fixed-target stopping rule above extends to an adaptive provenance certificate in Lemma 5.6.

Usable attribution is computed from available semantic metadata. Original generation and encapsulation slots belong to a fixed lifetime budget $[G] \times [C]$: a fresh draw receives a new slot, while copies retain their source. Each candidate induces its own protected records and report-key closure. Proposition 5.8 decomposes the score in the native experiment before comparing those local continuations. It retains both unattributable eligible executions and permitted key projections as explicit native-game terms.

The computational composition evaluates accepted operations, eligibility, and all public outputs anew in each comparison world. Exact component repeats retain one response and one correctness event. The native constructor map in Section 5.4 identifies

the storage operations used by these interfaces; Section 8 specifies the joint root and message continuation to which their bounds connect.

5 Computational interfaces and composition

The origin model provides the accounting boundary for a computational experiment. Each original generation or encapsulation invocation receives a slot when its entropy is reserved, including attempts that later fail. Replaying or copying that reservation retains its slot. Fix public lifetime budgets $G, C \geq 1$ and write $J = [G] \times [C]$. Unused slots remain unused. These proof-resource budgets differ from a bound on the number of records resident at one time.

The native application experiment evaluates the selected target after the adversary's interaction. Eligibility requires both original reservations to follow the selected cut, secrecy of the actual target signing keys and target-epoch entropy, and no acquisition of a selected-session version saved after that cut. It admits the asynchronous transitions of the construction, including sending after the local FinalReady transition.

We first construct the private-record and immutable-report interfaces. We then connect their candidate-specific continuations to one native score, using original provenance to certify permanent local stopping. After the native constructor correspondence, a joint key-schedule theorem derives the selected first-message continuation from a uniform recovered contribution. The common-virtual-game accounting variant is retained in Appendix D.

5.1 An explicit renderer for private images and intents

The alias lemma preserves a renderer that can read every original payload. A component reduction additionally needs to execute record management when the selected KEM challenger supplies public outputs and a shared-secret challenge without supplying the selected random tape. We construct that record-management algorithm for a finite image/intent interface. Its tables contain immutable handles, known fields and locally sampled sealing coins. The protocol's public computations are supplied by a common functional interface F, instantiated by the original-operation adapter. This section holds F fixed and isolates the image/intent contribution to the record connection in Proposition 5.8.

Typed record interface.

An opaque block is an immutable fixed-length private value, represented by a handle and its known length. Blocks may be correlated and may themselves be retained sealed reservations. A private image recipe contains at least one such block; an atom-free recipe is fully known. Local image-encryption and intent-authentication keys are available to the simulator. Fix a public budget M for all fresh image-sealing invocations, including failed attempts and atom-free images. Exact copies and exact retries retain their invocation. The following operations specify the interface.

1. *Seal.* Before sampling its nonce, invocation i fixes the recipe, its length and the non-nonce header fields. It samples independent $N_i \leftarrow \{0,1\}^n$ and privately

constructs

$$B_i = \mathsf{Hdr}_i \| N_i \| C_i,$$

where C_i is the specified deterministic local authenticated encryption under that nonce. Local encryption and intent authentication use their prescribed fixed algorithms without additional shared-SHA3 calls. All admitted image formats place the nonce at the same offset. The reply consists of the image handle, length and declared non-nonce metadata; N_i, C_i and the recipe's private blocks remain private.

2. *Intent.* A complete canonical prefix P fixes the tag, binding, predecessor and the image-head value used in this execution. Its tag determines its length. The intent bytes are

$$W_{i,P} = P \| B_i \| \mathsf{MAC}_{k_i}(P \| B_i),$$

with one fixed authentication key k_i for image i. An exact repeat retains the same (i, P) entry. Image and intent tags are disjoint.

3. *Public commitments.* For a designated private image, the ideal-commitment experiment returns independent uniform 256-bit values cached by i for its head and by (d, i, P) for its intent digests. The domain d ranges over the five admitted intent-digest domains. Atom-free objects retain their real hashes and ordinary raw-byte outputs. Later validation and zero-value checks use the actual replies in that execution.

4. *Equality and storage.* Equality queries compare complete retained image or intent handles. Copy, retain, persist, delete, crash and resume update an explicit durable-state directory. Storage choices, errors and branch decisions depend on available metadata, lengths, commitments and F outputs. A failure after commit preserves that commit.

5. *Acquisition.* Atom-free objects are rendered from their known inputs. For an opaque object, the admitted successful acquisition emits a computable event that permanently falsifies the fixed-candidate eligibility before its raw bytes are returned. The stopped score is zero, as in Corollary 4.8. Unavailable-object requests retain their ordinary error result.

All other public computation uses these replies, known inputs, F, and the shared public oracles. This is a classical record interface; the adversary may have quantum private state and quantum access to those public oracles. The successful-acquisition condition is supplied by the recovery experiment's existing event rule. For a fixed target session s and cut c, acquisition of an original version v with $\mathsf{subject}(v) = s$ and $c < \mathsf{save}(v)$ supplies such an event. Its original-record provenance determines that save time, including when a later aggregate carries the same version. Lemma 5.6 gives a provenance certificate valid before the final cut is chosen. A separately permitted projection is described by its own output operation.

Table algorithm.

The simulator stores an image table indexed by fresh sealing invocation, an intent table indexed by image and complete prefix, a commitment cache, and the directory with its provenance and acquisition ledgers. Each new image entry records the recipe handles,

length, known fields and a sampled nonce. Each new intent entry records (i, P) and its known length. The simulator samples designated-private commitments on first use and retrieves them on exact retry; it constructs and hashes atom-free objects normally. Equality is structural: images compare equal exactly when their indices match, and intents compare equal exactly when their pairs (i, P) match. The storage directory determines the state recovered after a crash, including an acknowledged or uncertain write that actually committed. At the stipulated permanent-failure acquisition event the computation returns zero. Every other transition executes the common public code and F. Thus the algorithm uses no selected opaque block bytes.

Lemma 5.1 (Structural equality of complete records) *If the nonces of fresh image invocations are pairwise distinct, then*

$$B_i = B_j \iff i = j, \qquad W_{i,P} = W_{j,P'} \iff (i, P) = (j, P').$$

Proof One immutable image handle retains one byte string. Two different image invocations have different nonces at the same explicit offset, so their complete wires differ. Intent tags first separate different formats. For one tag the prefix length is fixed: unequal prefixes differ before the target image, while equal prefixes with different images differ at the target's nonce. The same image and prefix give the same deterministic MAC under the fixed key. Disjoint tags also separate images from intents. This argument uses neither distinct private block values nor an injectivity assumption about the encryption algorithm. □

Let I be the materialized ideal-commitment experiment, which retains real private bytes and answers equality by comparing them. Let T be the table algorithm above. Both use the stipulated fixed-candidate stopping rule and the same public-oracle model and interface F.

Theorem 5.2 (Opaque image/intent simulation) *For at most M fresh seals with independent n-bit nonces, define*

$$\delta_{\text{nonce}} = \min\{1, \tbinom{M}{2} 2^{-n}\}.$$

For every efficient terminal bit D computed from the public view, available metadata and event ledger, F outputs and its own independent test coins (including its retained application challenge bit),

$$|\Pr[I : D = 1] - \Pr[T : D = 1]| \le \delta_{\text{nonce}}.$$

For a score $Z \in [-1, 1]$ computed from the same available inputs, $|\mathbb{E}_I Z - \mathbb{E}_T Z| \le 2\delta_{\text{nonce}}$.

Proof Couple the fresh sealing nonces, private-commitment replies, scheduler choices, the state of F, and the public-oracle transformations. Before the first nonce collision, maintain agreement of the directories, known fields, immutable references, public bytes, statuses and semantic ledgers. Sealing adds the same metadata and nonce to corresponding entries; its private ciphertext is confined to the materialized entry. Intent creation adds the same complete prefix and target reference. A fresh private commitment gets the coupled reply, and a repeat consults the same cache. Atom-free commitments and acquisitions use identical materialized inputs. Lemma 5.1 gives agreement of equality replies. The common public computation

consequently makes the same transitions and oracle calls. Copying, committed or failed writes, deletion, crash and resume preserve the directory invariant. An opaque acquisition emits the same permanent-failure event before either computation needs to return its unknown bytes, and both stopped scores are zero.

The next nonce is independent uniform even when the decision to seal is adaptive. Across at most M invocations, a union bound charges $\binom{M}{2}2^{-n}$ for any collision. Coupled terminal bits differ only on that event; bounded scores can differ by at most two. Identical classical replies and shared-oracle transformations also preserve a quantum adversary's state at every coupled good prefix. The argument uses no measurement of a quantum query. □

Connecting real commitments.

After replacing complete-wire equality by structural equality, let ϵ_{com} bound the bit-distinguishing gap between real private record digests and the independently cached commitments of the typed interface. Both games run the same public code, ordinary known-record computations, available metadata and declared stopping rule. The comparison counts the full distinguishing computation against their shared public oracle. It is a cryptographic input to the following composition, separate from the explicit table simulation. Appendix B records the concrete multi-record bound under fixed-input injectivity, per-nonce confinement and the prescribed local encryption/authentication algorithms.

Corollary 5.3 (Real-record to table-algorithm bound) *For the declared fixed-candidate interface and the commitment conditions above, let R be the real-digest materialized experiment. With $n = 192$,*

$$|\Pr[R : D = 1] - \Pr[T : D = 1]| \leq \min\{1, \epsilon_{\mathrm{com}} + \delta_{\mathrm{nonce}}\}.$$

The corresponding $[-1, 1]$ score bound is $\min\{2, 2(\epsilon_{\mathrm{com}} + \delta_{\mathrm{nonce}})\}$. When the interface exposes no private-wire equality query, its additional δ_{nonce} term is zero.

Proof First replace real byte equality by structural equality, coupling exactly as above and charging δ_{nonce}. The resulting game has no extra nonce-dependent equality observation, so apply the stated private-commitment replacement there. Its ideal form is implemented exactly by the table algorithm. A triangle inequality proves the bit bound. Applying that bound to a test that returns one with probability $(1 + Z)/2$ proves the score bound. Without a private-wire equality operation the first replacement is absent. □

The hybrid order starts with constructible real record inputs, replaces their private commitments, and then uses the table algorithm for the KEM challenge embedding. Every changed commitment is consumed by that world's subsequent intent prefixes, checks and outputs. For an adaptive origin-pair event E_j, early stopping is justified only by a computable event making E_j false for every continuation and every still-admissible final target choice attributed to j. The disjoint-event accounting above retains this per-pair obligation.

5.2 Immutable report families and reverse-order composition

A later cleanup report can authenticate an earlier report's private key and tag. We compose the one-report HMAC replacement by creation order, retaining these dependencies. The reduction below supplies a concrete algorithm for handling a selected unknown report key: later dependent rows are already ideal, and earlier immutable transcripts precede that key.

Finite interface.

Fix a public budget N for fresh report-key draws and index the actual draws chronologically. Failed attempts retain their indices; a new draw after definite absence receives a new index. Exact replay of a retained original report keeps its index, key, and transcript. Each K_i is an independent uniform 256-bit value, with no rejection of the zero key.

Let $\mathcal{X}$ be the source-atom handles whose bytes the subsequent KEM adapter will not possess. In this report replacement, their real bytes are still available to the surrounding source experiment. Before using K_i, the machine fixes a bounded immutable transcript recipe m_i. It can contain available literals and protocol outputs, atoms in $\mathcal{X}$, earlier raw keys K_j with $j < i$, and previously returned tags. Its own key and tag do not occur in m_i. Lengths, dependency sets, and admission errors are computable from the recipe metadata. Native code may draw the key before assembling the transcript, preserving that order and its failures, provided those inputs do not depend on the draw. Every direct or computed use of a raw key retains its origin dependency; returned tag bytes are available values.

Compute an online protection flag before using the fresh key:

$$P_i = \mathbf{1}[\mathrm{deps}_{\mathcal{X}}(m_i) \neq \varnothing \ \vee \ \exists j < i : \ P_j = 1 \ \wedge \ K_j \in \mathrm{deps}_K(m_i)] \,. \tag{8}$$

This is forward closure under raw-key dependencies. It need not propagate through a previously returned tag. The flag uses origin references, not actual key equality or the primitive challenger's bit, and is recomputed in each comparison world's history.

The following operations specify the interface. Creation computes $t_i = \mathrm{HMAC}_h(K_i, m_i)$ and retains the actual nonzero-ID check, encoding errors, and persistence outcomes. Replay or validation names the original report and uses that same m_i; supplied identifiers can be compared with its retained tag. This operation does not submit changed private bytes as a new transcript under K_i. Acknowledgement, terminal-tag comparison, copying, and crash recovery preserve their ordinary original-handle semantics. Private storage uses the explicit recipe tables of the opaque-record interface, including its already idealized commitments and its separate error accounting. Public report bodies and other outputs use available inputs or their specified application interface. Protected raw keys have no additional unlisted computation or observation channel.

The full real experiment G retains each declared fragment/key disclosure. Let $\mathsf{Bad}_{\mathrm{rep}}$ be the first successful outward projection whose recipe depends on a raw K_i with $P_i = 1$, detected by its actual origin map before returning the dependent bytes. An unavailable fragment, a terminal tag alone, or a projection without that dependency does not trigger it. Define G^{stop} to end with score zero at this event. This is an analysis

game; a report-key fragment is not thereby declared to be a selected-session acquisition or a new freshness failure.

One-row bound and hybrids.

Use one public compression oracle h for all rows and quantum queries. The one-row experiment draws a uniform hidden 256-bit key K, independently of h and oracle-independent initial auxiliary state. The algorithm performs its complete counted preparation and chooses one classical transcript m of length at most L without a raw-key input. Its first row query returns either $\mathrm{HMAC}_h(K, m)$ or an independent uniform 256-bit tag; at most $T \geq 1$ row calls repeat this same transcript and cached answer. The algorithm has no additional raw-key, midstate or key-dependent output interface. It may continue querying the same h and output a bit. The budget q counts all public compression queries, including preparation, other real rows, continuation and the terminal test. Independently sampled known keys may coincide with K; their ordinary calls to h remain allowed and counted.

Let $\epsilon_{\mathrm{row}}(T, q, L)$ be any upper bound on this experiment's bit-distinguishing gap. This operational definition is the primitive input to the composition theorem. Appendix B gives the compression-random-oracle instantiation, its padding convention and query accounting. A zero answer is retained for the protocol's actual error check.

Define H_k, for $0 \leq k \leq N$, by running the stopped experiment with every unprotected row real, every protected row $i \leq k$ real, and every protected row $i > k$ replaced by its own independent uniform cached tag. All later recipes, flags, errors, and selections use that world's actual replies. Thus $H_N = G^{\mathrm{stop}}$ and $H_0 =: S^{\mathrm{stop}}$. A zero ideal tag takes the original error branch; it is never resampled. An absent invocation makes no row query.

Theorem 5.4 (Reverse-order report composition) *For each slot i, let L_i, T_i, q_i uniformly bound the transcript length, row calls, and complete ordinary compression work of the reduction below, including its prefix, future computation, other real rows, and final test. Set*

$$\epsilon_{\mathrm{rep},i} = \epsilon_{\mathrm{row}}(T_i, q_i, L_i), \qquad \epsilon_{\mathrm{rep},\Sigma} = \min\left\{1, \sum_{i=1}^{N} \epsilon_{\mathrm{rep},i}\right\}.$$

Under the stated interface, every common efficient terminal bit test D satisfies

$$\left|\Pr[G^{\mathrm{stop}} : D = 1] - \Pr[S^{\mathrm{stop}} : D = 1]\right| \leq \epsilon_{\mathrm{rep},\Sigma}.$$

For a terminal score $Z \in [-1, 1]$ which is zero on stopping,

$$|\mathbb{E}_{G^{\mathrm{stop}}} Z - \mathbb{E}_{S^{\mathrm{stop}}} Z| \leq 2\epsilon_{\mathrm{rep},\Sigma}.$$

The tests use the common available view, semantic ledger, protocol-interface outputs, and independent test coins, including the retained application challenge bit. They have no extra access to protected key or transcript bytes.

Proof Fix i and compare H_i with H_{i-1}. Run the full prefix from setup, locally sampling all earlier keys and computing their real rows. Before the selected row is used, both hybrids have the same prefix rules. If slot i is absent or unprotected, their behavior agrees and no

row challenge is used. Otherwise install the one-row challenger's key as an opaque $\mathsf{KeyRef}(i)$. The transcript m_i is computable: it excludes its own key, all referenced earlier keys were locally sampled, and the source atoms are still available at this proof stage. Submit this actual transcript to the row oracle when its first MAC call is reached, retain its reply, and run the original nonzero and persistence checks.

Maintain the original-report, key-origin, recipe, durable-state, and event tables. Every available output agrees with the selected hybrid. Earlier real rows remain computable because their immutable transcripts predate K_i and their keys are known. A future unprotected row cannot depend on $\mathsf{KeyRef}(i)$: such a dependency would make it protected by (8). Compute that row with its locally sampled key and public h. A future protected row has index greater than i and is already ideal in both hybrids; sample its first tag, cache it by original invocation, and retain its symbolic recipe without evaluating the unavailable bytes. This includes a later report which contains K_i or another earlier protected key and tag. Flags and subsequent known fields are recomputed using the actual returned tags.

For replay and validation, the selected row is called only on its original m_i; preserve each counted recomputation and compare supplied identifiers with its retained reply. Copying, failure after a durable commit, and crash/resume use the same immutable handles and the actual reached storage state. They neither redraw a key nor replace a transcript. Public decisions use available metadata and common-interface outputs. Before an outward protected-key projection, the same ledger event $\mathsf{Bad}_{\mathrm{rep}}$ enters the common stopped state with application score zero; any allowed terminal-bit postprocessing uses that state. No unavailable key bytes are serialized. Unprotected-key fragments retain their known bytes. This gives an explicit online simulation throughout the continuation.

The selected real row therefore produces H_i, and its ideal counterpart produces H_{i-1}. All other real keys are sampled as actual independent byte strings. If one happens to equal the unknown challenged key, evaluating its real MAC with public h automatically preserves that equality; those queries belong in q_i. No comparison with the hidden key or distinct-key conditioning is used. Intentional aliases of the same original report instead share its single row cache.

The reduction executes the adversary's quantum queries to the same h and introduces no measurement of those queries or oracle-dependent initial advice. The one-row lemma bounds the adjacent bit difference by $\epsilon_{\mathrm{rep},i}$. Sum over the reverse-order hybrids and cap at one. For the score, use the allowed randomized terminal test which outputs one with probability $(1+Z)/2$; its difference is half the score-expectation difference, giving the displayed factor two. □

Corollary 5.5 (Original eligibility and real disclosures) *For the full real experiment G, define*

$$\eta_{\mathrm{rep}}(Z) = \mathbb{E}_G\left[|Z|\,\mathbf{1}_{\mathsf{Bad}_{\mathrm{rep}}}\right].$$

Then

$$|\mathbb{E}_G Z - \mathbb{E}_{S^{\mathrm{stop}}} Z| \le \min\left\{2, \eta_{\mathrm{rep}}(Z) + 2\epsilon_{\mathrm{rep},\Sigma}\right\},$$

and hence

$$|\mathbb{E}_G Z| \le \min\left\{1, \eta_{\mathrm{rep}}(Z) + 2\epsilon_{\mathrm{rep},\Sigma} + |\mathbb{E}_{S^{\mathrm{stop}}} Z|\right\}.$$

For $Z = \mathbf{1}_{\mathsf{Eligible}}(-1)^{b\oplus b'}$, $\eta_{\mathrm{rep}}(Z) = \Pr_G[\mathsf{Eligible} \wedge \mathsf{Bad}_{\mathrm{rep}}]$.

Proof Couple G and G^{stop} up to the first $\mathsf{Bad}_{\mathrm{rep}}$. On its complement the runs agree; on that event the stopped score is zero while G continues with the actual disclosed bytes. Therefore

$$\mathbb{E}_G Z - \mathbb{E}_{G^{\mathrm{stop}}} Z = \mathbb{E}_G[Z\mathbf{1}_{\mathsf{Bad}_{\mathrm{rep}}}].$$

Bound its magnitude by $\eta_{\mathrm{rep}}(Z)$ and apply Theorem 5.4. No conditioning on eligibility or independence of later target selection is used. □

Consequences for the record connection.

The residual is zero on a disclosure surface whose actual provenance already makes the original target permanently ineligible. A batch-only report-key fragment requires its own mapping: it is not automatically a member-session capture. This distinction is material. Given a known transcript and its later disclosed key, recomputing HMAC distinguishes the real tag from an independent tag with difference $1 - 2^{-256}$. This example does not assert that a native batch fragment also reveals its members' complete transcript.

The report replacement precedes KEM embedding. Afterwards every protected row needs only its original cached tag, and every unprotected recipe is free of $\mathcal{X}$ and protected raw-key dependencies. The report layer can therefore use opaque source handles for tag generation and immutable tag validation. Direct source-atom or application-plaintext outputs retain their own adapters and acquisition rules. The preceding private-record hop keeps its own error and acquisition correspondence; this theorem does not infer that correspondence from the report stopping game. One precise native composition first charges the actual eligible-exposure residual, then abstracts the stopped native experiment, and finally applies the report hybrids. The record correspondence must be proved for that explicit stopped interface; a key-only fragment does not inherit the original renderer's permanent-target-failure premise. Likewise, arbitrary changed-byte import is a separate MAC-query interface from the immutable validation proved here. These are specific interface boundaries of the composition, while the dependent immutable rows themselves are covered by the explicit reduction above.

5.3 Candidate-local simulation in the native experiment

A guessed origin pair can determine which records are opaque, which report rows are protected, and where the reduction stops. We therefore localize the score in one native experiment before comparing its candidate-specific continuations. This formulation preserves the original target eligibility and does not require the local real-component games to coincide.

Let $\mathsf{G}_{\mathrm{nat}}$ be the full native experiment and $J = [G] \times [C]$ its fixed reservation-slot budget. Write E for original eligibility, A for usable unique attribution, and $I \in J$ for the attributed pair on A. These variables describe the actual accepted logical challenge send; selection does not transfer its test bit to another send. For a bounded application score $S \in [-1, 1]$, set

$$Z = \mathbf{1}_E S, \qquad Z_j = \mathbf{1}_{E \wedge A \wedge (I=j)} S.$$

For each j, let U_j and B_j be efficient, cumulative prefix events computed from the available classical provenance ledger. They are noninterfering monitors: before stopping they change no native response or protocol transition and do not inspect an unknown tape or a quantum query. A *free stop* U_j permanently falsifies $E \wedge A \wedge (I = j)$ for every continuation. A *charged stop* B_j may instead be a permitted protected-report-key projection. Define $\mathsf{G}_{\mathrm{nat}}^{[j]}$ to run the native code until the first such event, then enter a

stopped state with candidate score zero before returning the monitored bytes. An event for one candidate does not stop the original game or another candidate's reduction.

Lemma 5.6 (Origin-anchored adaptive stopping) *Fix candidate j and one of its original reservation slots o which has occurred. Suppose a successful capture supplies an intact version v with*

$$\mathsf{subject}(v) = \mathsf{session}(o), \qquad \mathsf{res}(o) \le \mathsf{save}(v).$$

Assume usable attribution to j assigns its member origins to the selected session, and eligibility places the cut before both reservations. Then this capture is a free U_j event, even if the other reservation or challenge send has not occurred and the final cut has not been selected.

Proof In any continuation with $E \wedge A \wedge (I = j)$, the selected session is $s = \mathsf{session}(o)$ and the selected cut satisfies $c < \mathsf{res}(o) \le \mathsf{save}(v)$. Acquisition has already inserted v into the cumulative archive. Thus the continuation has acquired a version of s saved after c, contradicting the original acquisition conjunct. Copies, resealing, deletion, and recovery preserve this contradiction because the subject, original save event, and acquisition are immutable. If the other reservation or challenge send never occurs, the attributed eligible event is already false. Decryption and a final cut value are unnecessary for this certificate. □

Actual retention of o in v supplies the time inequality by reservation/save causality; an independently recorded later save of the same session also suffices. More generally, let L_s be the largest original save time of an acquired intact version of session s, with $L_s = -\infty$ for an empty set. For an existing pair (g, a), the original temporal eligibility conditions are exactly

$$L_s \le c < \min\{\mathsf{res}(g), \mathsf{res}(a)\}.$$

The remaining eligibility conditions retain their own tests. This statistic uses original version times, not the enclosing image's latest sealing time. A batch-only key fragment or a public commitment supplies no intact-session witness merely because a report depends on that session.

One native residual.

In the full native run let B_I denote the charged event for its final attributed pair, used only on A, and define

$$\eta^{\mathrm{nat}}_{\mathrm{attr}} = \Pr_{\mathsf{G}_{\mathrm{nat}}}[E \wedge \neg A], \qquad \eta^{\mathrm{nat}}_{\mathrm{proj}} = \Pr_{\mathsf{G}_{\mathrm{nat}}}[E \wedge A \wedge B_I].$$

These are disjoint events in one game. In particular,

$$\eta^{\mathrm{nat}}_{\mathrm{proj}} = \sum_{j \in J} \Pr_{\mathsf{G}_{\mathrm{nat}}}[E \wedge A \wedge (I = j) \wedge B_j].$$

A projection affecting only another candidate is not charged to the selected one. For the report construction, B_j is the actual protected-key projection event for the closure induced by that candidate's source family. Its probability is retained explicitly; the theorem does not declare it negligible or turn the fragment into a session snapshot.

Lemma 5.7 (Native score localization with local stops) *Let Ψ_j be the same terminal code for $\mathbf{1}_{E\wedge A\wedge(I=j)}S$ in each experiment, with value zero in its stopped state. Then*

$$\mathbb{E}_{\mathsf{G}_{\mathrm{nat}}^{[j]}}\Psi_j = \mathbb{E}_{\mathsf{G}_{\mathrm{nat}}}[Z_j\mathbf{1}_{\neg B_j}], \tag{9}$$

$$\mathbb{E}_{\mathsf{G}_{\mathrm{nat}}}Z = \mathbb{E}_{\mathsf{G}_{\mathrm{nat}}}[Z\mathbf{1}_{\neg A}] + \mathbb{E}_{\mathsf{G}_{\mathrm{nat}}}[Z\mathbf{1}_{A\wedge B_I}] + \sum_{j\in J}\mathbb{E}_{\mathsf{G}_{\mathrm{nat}}^{[j]}}\Psi_j. \tag{10}$$

Proof Until its first local stop, the stopped machine has the native prefix distribution. On a run with B_j, its candidate score is zero. On a run with U_j, every native continuation already has $Z_j = 0$, whether or not B_j later occurs. On all other runs the executions and candidate scores agree. This proves (9). Partition the one native score by attribution and its unique index; pointwise, $\sum_j Z_j\mathbf{1}_{B_j} = Z\mathbf{1}_{A\wedge B_I}$. Substitution proves (10). The final index is used in this identity, not predicted by the online monitor. For a quantum adversary the argument compares identical classical-prefix instruments and their stopped continuations; it neither clones a state nor measures an oracle query. □

Local cryptographic comparisons.

For each j, let $\mathsf{V}^{[j]}$ be its specified real-component continuation after the local source/record/report replacements, and let $\widetilde{\mathsf{V}}^{[j]}$ be its corresponding random-component game. All terminal events are recomputed in their own game by the same score code; no event from a native transcript is transferred to an altered transcript. Suppose the actual stopped-interface correspondence gives

$$\left|\mathbb{E}_{\mathsf{G}_{\mathrm{nat}}^{[j]}}\Psi_j - \mathbb{E}_{\mathsf{V}^{[j]}}\Psi_j\right| \le \rho_j.$$

This is a local score bound, including its native source correspondence. A binary record or report bound contributes twice its value to this score comparison. The stopping certificate does not itself establish the correspondence for permitted partial projections.

For the remaining component step, specialize to $S \in \{-1,+1\}$, so that $\Psi_j \in \{-1,0,+1\}$. Suppose its actual fixed-pair adapter satisfies, for $\sigma \in \{-1,+1\}$,

$$\left|\Pr_{\mathsf{V}^{[j]}}[\Psi_j = \sigma] - \Pr_{\widetilde{\mathsf{V}}^{[j]}}[\Psi_j = \sigma]\right| \le a_j^{\sigma} + \delta_j.$$

Here a_j^{σ} includes the complete derived distinguisher and its shared oracle work; δ_j is the selected pair's correctness term. Write $b_j = \mathbb{E}_{\widetilde{\mathsf{V}}^{[j]}}\Psi_j$.

Proposition 5.8 (Candidate-local native accounting) *Under the stated local comparisons, the native message-guessing advantage satisfies*

$$\mathrm{Adv}_{\mathrm{msg}} \le \min\left\{\frac{1}{2}, \frac{1}{2}\left(\eta_{\mathrm{attr}}^{\mathrm{nat}} + \eta_{\mathrm{proj}}^{\mathrm{nat}} + \sum_{j\in J}\left(\rho_j + a_j^{+} + a_j^{-} + 2\delta_j + |b_j|\right)\right)\right\}.$$

Proof The first two terms of (10) have magnitudes at most the two native residuals. The local correspondence costs ρ_j. Subtracting the positive- and negative-event comparisons bounds the local component score change by $a_j^+ + a_j^- + 2\delta_j$. Sum the candidate terms, apply the triangle inequality, and use $|Z| \le 1$. Completing invalid runs by an independent fair guess gives $\mathrm{Adv}_{\mathrm{msg}} = \frac{1}{2}|\mathbb{E}_{\mathsf{G}_{\mathrm{nat}}} Z|$. □

The ideal continuations may have different adaptive selectors. The terms $|b_j|$ therefore remain separate; they are not probabilities from one partitioned ideal world. If a genuine common intermediate game and a global comparison have separately been established, that shared error can instead be charged once before the common-game accounting. The proposition above covers the candidate-local constructions without asserting such a global identification.

5.4 Native constructors and retained-object transitions

We map the declared record constructors to the fixed Continuity source. The accompanying audit contains 37 entries across 27 directly cited files, with exact source ranges and checked Git-blob and SHA-256 identities. It separates constructor layout, state transition, known-input computation and the protocol's additional output interfaces. The following correspondences give the native instances of the objects used in the preceding proofs.

Images, intents and commitments.

`durable::seal` encodes and bounds the payload before drawing a 24-byte nonce. The four image tags place that nonce at byte offsets $[80, 104)$, followed by the XChaCha20-Poly1305 ciphertext and authentication tag. The target-bearing `PendingWrite` formats have a canonical tag-dependent prefix, the complete original target image, and a deterministic HMAC under the journal-derived intent key. These layouts supply the fixed offset, prefix and intact-target premises of Lemma 5.1. The wrapping key for a reserved operation and the intent-authentication key use their respective derivation domains; the simulator can retain these local storage keys independently of the selected KEM random tape.

The source uses one image-head domain and five target-bearing intent-digest domains. Their callers feed the actual returned digests into fences, checkpoints and subsequent validation. Cancellation intents carry known descriptors instead of a private target image; selection rows and public archive descriptors likewise use their actual known-input computations. This variant distinction supplies the real-versus-designated-private split used by the commitment table.

Reservation and durable identity.

Both generation and encapsulation paths persist the reserved operation token before invoking the corresponding KEM computation. An original retry opens that token and consumes its retained entropy. The write-intent path first retains the complete pending wire. Applying it compares the retained original wire, checks the expected and next heads, installs its original target bytes, and removes the pending entry. An already installed exact target is handled as the same operation. A sibling-session write

Family	Immutable transcript and key placement	Ordinary report result
Session closure	The report MAC covers the session payload with its own 64-byte report/key tail zeroed. The pending key is retained in that tail. Earlier resolution keys can occur in the preceding traffic entries.	Session and epoch progress, counts and loss metadata.
Fanout abandonment	The MAC covers batch metadata and complete member payloads. Its own key is held in a separate batch tail, outside that metadata. Member payloads can include earlier resolution keys.	Per-member progress and loss metadata.
Closed-epoch resolution	The MAC covers the old epoch's counters, message identifiers, retained digests and skipped indices, excluding its own key. Its pending key is retained in the traffic entry.	Includes unconsumed plaintext for the authorized application.

Table 3 Source correspondence of the report families. The application's authorized plaintext output retains its own audience in the security experiment. Report-key fragments use their actual record content and origin mapping.

can instead seal a new image containing unchanged target payload bytes, preserving the target's source while creating a new physical sealing invocation.

Uncertain commit and recovery.

The native state distinguishes a caller's outcome from the durable directory. An unconfirmed commit maps to `CommitUncertain`; the owner is closed and subsequently reopened against the retained image and pending target. An actual committed write therefore persists across an error response. The explicit storage machine records this outcome and retains the original operation on resumption. The cited acquisition rule uses the contained session version's original save event, including when that version is carried by a later aggregate.

Report dependencies.

Table 3 identifies the three report families. The resolution key and pending tag are serialized in an epoch's traffic entry. A later closure or fanout report can authenticate that complete entry. This gives the causal raw-key edge used by the reverse-order report argument. The code's exact retry paths retain an original key and tag; a genuinely new attempt draws a new key. The nonzero check applies to the returned identifier, with a real zero-ID failure branch. Pending keys use the full 256-bit space.

The normal retained-object path reopens a registered image and revalidates its original semantic state. The interface mapping assigns those validation calls to known state, the protocol interface F, or the immutable report row. Arbitrary modified storage inputs and separately returned projections receive their corresponding parse, authentication and output operations in an extended experiment. In particular, the batch-key record and a member's session payload have different contents: their acquisition events are determined separately. These typed boundaries preserve the existing target predicate while making each native call's role in the reduction explicit.

5.5 Joint continuation from an epoch root to a message

A recovered root feeds confirmation, traffic derivation and the next epoch. We treat these uses jointly and then isolate a selected message key. The resulting continuation permits later contributions, successor roots and the next chain seed to be disclosed. The current root is an internal value: all outputs are regenerated in each hybrid, rather than fixed before a root replacement.

Primitive experiments.

Let $F(k, m)$ be HMAC-SHA256 over one public quantum compression oracle $h : \{0,1\}^{512} \times \{0,1\}^{256} \to \{0,1\}^{256}$, in message-block, chaining-state order. All keys below have 32 bytes. Put

$$\Gamma(a) = \big(h(I(a), \mathrm{IV}), h(O(a), \mathrm{IV})\big),$$

where $I(a) = (a \| 0^{32}) \oplus \mathtt{36}^{64}$ and $O(a) = (a \| 0^{32}) \oplus \mathtt{5c}^{64}$. Initial auxiliary state is independent of h and the designated hidden uniform key. All oracle-dependent preparation is executed and counted, together with the complete continuation.

Write $\varepsilon_{\mathrm{Hjoint}}(C, T, q, L)$ for an output-bit gap bound between

$$\mathrm{Col}(a) = F(a, R), \quad \mathrm{Row}(m) = F(R, m) \quad \text{and} \quad V(\Gamma(a)),\ U(m),$$

where R is hidden uniform, U, V are independent private random functions, there are at most C classical column calls, T classical row calls of length at most L, and q ordinary compression queries, which may be quantum. Equal inputs are cached. Distinct salts with equal Γ share their column reply. The ideal here is the independent-table ideal, without a separately imposed cross-pair repair.

Write $\varepsilon_{\mathrm{Hrow}}(T, q, L)$ for the corresponding real-HMAC-row versus private-random-function bound for one uniform hidden key and *multiple adaptive messages*. This differs from the immutable report row experiment. Finally, $\varepsilon_{\mathrm{Hcol}}(q)$ bounds one distinct zero-salt evaluation $F(0, S)$, for hidden uniform S, versus an independent uniform reply; exact repeats reuse it. The retained compression-QROM arguments give admissible bounds for these operational experiments.

The finite continuation.

The initial root R is sampled independently of h and the initial auxiliary state, before any use. It may be presampled and withheld during counted preparation, which chooses the public session identifier s and transcript inputs. The previous root and old wrapping keys may be known. The current root has only the uses

$$P_0 = F(0, R), \qquad P_s = F(s, R), \qquad P_x^+ = F(R, S_x^+). \tag{11}$$

There are at most t_+ distinct 32-byte successor contributions S_x^+. They are supplied by known local computation, may be disclosed, and may depend on the available view. The controller may compute and disclose

$$R_x^+ = \mathrm{Expand}(P_x^+, I_x^+, 32)$$

and run arbitrary known subsequent computation from these values with the same public oracle. Repeated arguments remain consistent.

Let D and D_M denote the actual 40-byte rekey and 43-byte message domains. The three confirmation information strings are

$$\begin{aligned} I_r &= D \| \texttt{responder-confirmation}, \\ I_f &= D \| \texttt{final-confirmation}, \\ I_a &= D \| \texttt{receipt-confirmation}. \end{aligned}$$

Their keys are $\mathrm{Expand}(P_0, I_d, 32)$. The controller uses these keys in the real MAC, verification and signing algorithms. At most $M \geq 1$ distinct traffic information strings have the native form

$$\begin{aligned} I_T = D \| &\texttt{epoch-traffic/HKDF-SHA256/ChaCha20Poly1305/} \\ &\| \mathit{context}_{32} \| \,\mathrm{BE64}(e) \| \mathit{transcriptDigest}_{32}. \end{aligned} \tag{12}$$

It may be chosen after the response confirmation and its signature: the native transcript digest hashes the complete offer and response wires. Every repeated complete I_T reuses its entry, regardless of endpoint or wire representation. At the first allocation of one new string I^*, mark that entry and a direction $d^* \in \{1, 2\}$ before any of its blocks are produced. The choice may use the available prefix; a previously allocated entry cannot be reclassified as fresh. Later equal-info aliases inherit the same protection. This is a profile for a designated fresh traffic allocation, without a cross-endpoint wire-uniqueness premise.

The four 32-byte output blocks are

$$(s_{I,1}, s_{I,2}, a_{I,1}, a_{I,2}) = \mathrm{Expand}(P_s, I, 128).$$

The selected seed $s^* = s_{I^*,d^*}$, designated before allocation, has only the selected first-step use

$$\begin{aligned} I_M &= D_M \| \texttt{chain} \| \,\mathrm{BE64}(0), \\ P_M &= F(0, s^*), \qquad (s', K) = \mathrm{Expand}(P_M, I_M, 64). \end{aligned} \tag{13}$$

The opposite seed, confirmation keys, ACK keys, blocks at every distinct nonmarked information string and next seed s' may be disclosed. A repeated I^* remains the same protected entry. Real ACK computation and arbitrary candidate-ACK verification remain available. Subsequent chain computations use s' honestly. The selected key K has only its declared application encryption/decryption use. Current-root, PRK, old selected-seed and message-key bytes have no separate public return, comparison or hashing interface. Private persistence uses an already justified opaque interface; original candidate-local stops and projection accounting remain in force.

Lemma 5.9 (Adaptive information and fresh traffic blocks) *Replace each PRK-class HMAC row by an independent private random function, available only through HKDF-Expand. The*

resulting interface is exactly an independent block table indexed by the class, complete information string and block index, with consistent prefixes. This holds for adaptively chosen information strings. In particular, the first four traffic blocks are fresh at installation despite the confirmation dependence of I_T.

Proof For one private row W, the inputs are $I\|1$ and $T_{I,j-1}\|I\|j$ for $j \geq 2$, where each preceding block has exactly 32 bytes and the counter has one byte. Different counters separate inputs. For the same counter, deleting the counter and, when present, the fixed-width preceding block shows that equality requires equal I. This is true for every preceding output assignment. Equal (I, j) pairs retrieve equal inputs and replies inductively. New pairs therefore receive independent table blocks, even when I was chosen from prior outputs and public-oracle computation.

The confirmation information lengths are $62, 58, 60$ bytes and each traffic information length is 155, so their table entries differ even if the zero and session salts share a PRK. The marked I^* is also distinct from every earlier traffic entry. Importantly, this exact rewrite removes the internal parent-row inputs containing a preceding secret block. Because that row has no other interface, they leave no extra use of the selected traffic seed. Public quantum-oracle computation is unchanged. □

An available test uses the public view, event ledger, declared locally computable outputs and its independent test randomness. It cannot inspect the designated private roots, PRKs, seeds, message key or tables. For the application experiment below we can fix either challenge bit and then average; the test is not given a hidden primitive key.

Let G_R be this real schedule with uniform root. In G_{leaf} replace (11) by $V(\Gamma(\cdot))$ and $U(\cdot)$, replace each column-PRK Expand interface by Lemma 5.9's block table, and replace (13) by two independent cached blocks s', K. Real confirmation/ACK operations and successor-root derivations continue from the known replies; all signatures, checks and later outputs are regenerated within the current game.

Theorem 5.10 (A selected message leaf with joint continuation) *Let q_R, q_0, q_1, q_s, q_M bound the entire ordinary compression-query work of the five reductions below. Set*

$$\begin{aligned}\Delta_{\text{sched}} = \min\{1,\ & \varepsilon_{\text{Hjoint}}(2, t_+, q_R, 32) \\ & + \varepsilon_{\text{Hrow}}(3 + 4M, q_0, 188) + \varepsilon_{\text{Hrow}}(4M, q_1, 188) \\ & + \varepsilon_{\text{Hcol}}(q_s) + \varepsilon_{\text{Hrow}}(2, q_M, 89)\}.\end{aligned} \tag{14}$$

For every available terminal bit, the probability gap between G_R and G_{leaf} is at most Δ_{sched}. For a common available score in $[-1, 1]$, the expectation gap is at most $2\Delta_{\text{sched}}$. The second PRK-row term is omitted when the interface itself rules out a second class.

Proof First give every current-root column and row call to the joint primitive experiment. Their replies suffice to compute all Expands, confirmations, signatures, traffic and successor roots. In particular, disclosed successor contributions and roots require no raw current root. This gives the first term and retains the same public oracle throughout.

Next replace the at most two column-PRK rows in class-allocation order. Preallocate the zero-salt class privately if necessary; a second class exists only when $\Gamma(s) \neq \Gamma(0)$. For a designated class, install the row primitive's hidden uniform key as its first V reply, withheld

until allocation. The identifier is computed from known salts and h; it does not inspect the private reply. Every equal Γ reuses this handle. Answer Expands by querying the challenge row on the actual successive messages. Their output blocks are known, so real MACs, verifications, signatures and continuation can all be performed. Other class keys are sampled locally. All their real HMAC computations, and all preparation and future work, count as ordinary queries.

Three one-block confirmation Expands and at most M four-block traffic Expands give input lengths $63, 59, 61$ and $156, 188, 188, 188$. Thus the first class has at most $3 + 4M$ distinct row calls and a second class at most $4M$. A never-allocated class causes no challenge call. Their total is at most $3 + 4M$; the separate worst-case budgets avoid conditioning a primitive bound on the oracle-dependent class-equality event.

Independent V replies may coincide numerically. No inequality is assumed: known-key computations are performed through the same h, including accidental overlap with the challenge key, and are counted by the primitive experiment. An ideal row is the declared hybrid replacement. Intentional Γ aliases share that row in every game.

Apply Lemma 5.9 exactly. The selected traffic seed is now a fresh independent block, and its parent no longer exposes an input containing it. Presample the single-column challenge seed and install it at the newly marked entry before its blocks are produced. There is no earlier equal-info allocation; later aliases retain this protected handle. Its only raw use is $F(0, s^*)$; every other block is sampled locally. The next hop therefore makes P_M independently uniform at cost $\varepsilon_{\mathrm{Hcol}}(q_s)$.

Finally install the row challenge key at that fresh P_M. The two actual Expand inputs have lengths 57 and 89, giving the final term. The same exact block-table rewrite produces independent s', K. The next seed may be disclosed and used in all later chain computation; the reduction has every value it needs except the key reserved to the application interface. These five comparisons prove the bit bound. Randomizing a terminal bit with probability $(Z + 1)/2$ proves the score bound. Equivalently one may fix an independent application challenge bit, apply this bound, and average over that bit. □

The conclusion concerns independence *at key installation* with the continuation regenerated consistently. It does not assert independence of the key and a subsequently computed ciphertext, nor pseudorandomness of every intermediate root relative to a fixed complete future view.

Application confidentiality.

Use a one-key, one-encryption left-or-right AEAD experiment with nonce 0^{96}, equal-length messages, common associated data and the same additional public oracle h. Its classical decryption interface returns real plaintext or failure, except that an exact challenge query returns an authenticated marker. Let $\varepsilon_{\mathrm{AEAD}}$ be its absolute guessing advantage over $1/2$. An encryption-only profile uses the corresponding one-use privacy experiment.

The honest application creates the selected send. The network adversary receives ciphertext, metadata, status, ACKs and real authentication/error outcomes, and can request plaintext-free Resume(id). The actual payload and its retained send-intent digest are opaque application handles. Delivery and any payload-bearing resolution output are private to the honest application; the public continuation has no additional payload-dependent return. An exact challenge reception can therefore update public state using its authenticated marker and known metadata. Other ciphertexts use the declared decryption interface.

The send-origin normalization caches the encryption reply at its first *private computation*, even if that computation precedes durable commit or network release. Native crash recovery may repeat encryption, but always on the same original tuple; the reduction retrieves its cache. The cache includes the marked material's original allocation and authoritative owner/history, direction and index. Distinct material allocations are not merged merely because their epoch numbers agree.

Corollary 5.11 (First message of a recovered direction) *For an event A computable from the available view and ledger without reading the hidden AEAD challenge bit b, let $Z = \mathbf{1}_A(2\mathbf{1}[g = b] - 1)$. Under the application interface above,*

$$|\mathbb{E}_{G_R} Z| \le 2\Delta_{\mathrm{sched}} + 2\varepsilon_{\mathrm{AEAD}}.$$

Proof Theorem 5.10, applied for each fixed b and averaged, changes this score by at most $2\Delta_{\mathrm{sched}}$. In G_{leaf} give the selected encryption/decryption operations to the AEAD challenger and sample every other block locally. Confirmation, ACK, successor-root and subsequent-chain computation uses known material and the same public oracle. Cache the first encryption response; exact challenge reception needs only the marker and metadata. At the end return g when A holds and a fresh fair guess otherwise. Its centered success is exactly $\mathbb{E}Z$, bounded by $2\varepsilon_{\mathrm{AEAD}}$. □

The public application boundary is substantive. For two challenge plaintexts with different native send-intent digests, publishing the result of send_message(id, m_0, ad) after the challenge would distinguish a retained result from an intent conflict. No record acquisition is needed for that test. The network profile uses Resume while the honest application retains the plaintext-bearing API; original eligibility is unchanged.

Entry from a recovered contribution.

The uniform value above is the epoch root *after* root derivation. For a separately justified uniform combined contribution S, known previous root r and public response core, the actual derivation is

$$\begin{aligned} P = F(r, S), \qquad R = F(P, I_R \| \mathtt{01}), \\ I_R = D \| \mathtt{pending\text{-}root/HKDF\text{-}SHA256/} \\ \| \,\mathrm{hash}(\mathtt{response\text{-}core}, \mathit{core}). \end{aligned}$$

Its information string has 97 bytes and its sole Expand message has 98. Suppose this is one original Extract/Expand, with exact repeats cached, and S, P have no additional raw-use or disclosure channel. The retained root-initialization lemma applies with the entire controller above as continuation, at bit cost Δ_{init}. Its ideal reply can be presampled and withheld, providing precisely the joint uniform root initialization required here. It regenerates confirmations, signatures, traffic and future epochs; it does not replace a root inside a fixed public transcript.

Let $\mathsf{W}^{[j]}_{\mathrm{cont}}$ denote this uniform-combined-contribution continuation, with the same candidate-local score code Ψ_j, and put $b^{\mathrm{cont}}_j = \mathbb{E}_{\mathsf{W}^{[j]}_{\mathrm{cont}}} \Psi_j$. Consequently,

$$|b^{\mathrm{cont}}_j| \le 2\Delta_{\mathrm{init},j} + 2\Delta_{\mathrm{sched},j} + 2\varepsilon_{\mathrm{AEAD},j}.$$

The native accounting term $b_j = \mathbb{E}_{\widetilde{\mathsf{V}}^{[j]}} \Psi_j$ belongs instead to the random-component game, which may still use the real ContextBound combiner. If its actual complete comparison to $\mathsf{W}^{[j]}_{\mathrm{cont}}$ has bit gap at most $\varepsilon_{\mathrm{CB},j}$, then

$$|b_j - b^{\mathrm{cont}}_j| \le 2\varepsilon_{\mathrm{CB},j}, \qquad |b_j| \le 2\varepsilon_{\mathrm{CB},j} + |b^{\mathrm{cont}}_j|.$$

This comparison retains the controller, query interfaces and score/event code, counts all shared-oracle preparation and continuation, and recomputes events within each game's execution. Its actual tuple and disclosure conditions must hold. The randomized test with success probability $(\Psi_j + 1)/2$ gives the first inequality; the second is the triangle inequality. Native correspondence ρ_j, component and authentication terms, and original-game projection residuals remain in their established locations. This supplies the continuation bound for the specified profile, including known old roots and allowed later disclosure.

6 Mechanized recovery and separating executions

6.1 The multi-epoch symbolic theorem

The checked symbolic model includes inherited roots, alternating proposers, asymmetric confirmation, initial traffic, and unbounded epochs. Signing and wrapping keys may be shared across their modeled uses. Its Cut event exposes the selected wrapping keys; exposure of other epochs' entropy is represented separately. We state the main theorem in terms of its recorded events, preserving its original target and disclosure predicates.

Let τ be a trace of that model. Write $\mathsf{SessionKeys}(s, a, b)$ for the actual identity keys associated with session s, $\mathsf{KeyReserved}(s, e, P)$ and $\mathsf{EncapReserved}(s, e, Q)$ for the target reservations, and $\mathsf{Message}(s, e, r, k, m)$ for an application-message event. Define $\mathsf{AcquiredAfterSave}_c(s)$ to hold when there exist a role, an epoch, a revision, and matching Saved/Snapshot events for session s, with the *save* event strictly after c. The matching record's session projection is the relevant saved object.

Theorem 6.1 (Checked multi-epoch symbolic secrecy) *Suppose τ contains the corresponding SessionKeys, Cut at c, target reservation events at g and h, and* $\mathsf{Message}(s, e, r, k, m)$. *If*

$$c < g, \qquad c < h,$$

neither actual target identity key is disclosed anywhere in τ, no reserved entropy of target epoch e is disclosed, and $\mathsf{AcquiredAfterSave}_c(s)$ *is false, then τ contains no adversarial knowledge event* $\mathsf{K}(m)$.

The theorem was checked with all 24 reusable dependencies in one Tamarin process. An independent replay produced a byte-identical proof certificate. Appendix A

identifies the exact query, source, toolchain, replay receipt and complete auxiliary-query inventory.

The theorem uses the modeled cryptographic constructors and record interface. Its key scope follows actual identity ownership: a shared signer exposed through another session is still a disclosure of that target key. Likewise, a retained target projection saved after Cut remains subject to the acquisition condition when a sibling write carries it into a later aggregate image.

6.2 Executability and the freshness boundary

A finite two-round construction exercises the operational rules with an exposed ancestor root and satisfies the complete original positive existential formula. A separate source-derived evaluator checks its rule instances and global quantifiers. This provides a concrete execution for the modeled recovery premises.

The delayed-receipt construction instead contains a pre-Cut epoch-two generation reservation and a post-Cut encapsulation. It derives the target payload while satisfying the remaining main-theorem antecedents. Its point audit identifies $c < g$ as the sole false antecedent. Seven countertraces are rejected by the witness checker. Together, the positive and separating traces connect the recovery condition to executable schedules.

6.3 Native reservation disclosure

The native experiment stops a child process after committing a reservation, kills and reaps it, reopens the journal, and inspects the retained phase. It exercises two distinct boundaries:

- a 277-byte key-generation token, before key computation or offer publication;
- a 245-byte encapsulation token, after offer authentication and before encapsulation computation or response publication.

The predictor obtains the previous root, journal identity, wrapping key and captured token. Thereafter it uses the public policy configuration and later authenticated control wires. Its prediction interface has no journal or filesystem input, and the experiment withholds signer files and future private owners. A wrong wrapping key is rejected.

For each cut, both honest peers complete the rekey and reopen their journals. The driver then creates fresh 128-byte application payloads, sends three in each direction, and checks honest delivery. Independently reconstructed traffic owners authenticate and decrypt all six intercepted frames. The two cuts therefore yield twelve actual post-restart plaintext recoveries. The delayed-receipt extension repeats the six-message check at epoch two using one responder snapshot.

Proposition 6.2 (Execution-time and counter-based separation) *In the native executions, computing a KEM after restart, completing the target epoch, and subsequently reopening the journal do not make a previously disclosed reservation an independent secret contribution. Moreover,* $\min(c_A, c_B) + 1$ *is not a universal upper bound on exposed epochs.*

Table 4 Research questions and their validation objects. Native plaintext recovery, finite-model equality and component key derivation are recorded as separate outcomes.

Question	Design and observation
Which event fixes a contribution's origin?	Reservation-boundary captures, delayed receipts and sibling aggregate writes; reconstruction of subsequent traffic from the captured original inputs.
Which acquired objects jointly expose a source?	Iterative decryption and guarded routes on the same finite graphs, with explicit expected cases for key dependencies and later disclosures.
How does retained state interact with communication opportunities?	Pinned component snapshots across seven schedules, with reconstructed keys indexed by direction, epoch and message number.

Proof At both process-cut boundaries the captured reservation reconstructs the later honest KEM result exactly. Equations (1)–(2) then reconstruct traffic keys from the captured predecessor root and public wires; authenticated decryption verifies the six corresponding plaintexts after completion and restart. In the delayed-receipt execution, $(c_A, c_B) = (0, 1)$ at capture and six epoch-two plaintexts are recovered. Thus the proposed counter bound equals one while an exposed epoch equals two. □

6.4 An aggregate write preserves the target origin

A separate two-session native execution changes a sibling session and writes a new aggregate image. It checks exact equality of the retained target record, reopens and retries the original operation, and decrypts six subsequent target messages in each build profile. The new global revision and outer encryption nonce coexist with the old target reservation. This execution instantiates the alias/provenance distinction in the record model.

The native executions isolate the reservation and acquisition conditions used in Theorem 6.1. Section 5 gives the record simulations and common-experiment accounting for the corresponding computational interfaces.

7 Validation of exposure semantics

The validation asks how the formal exposure conditions appear in executable state and communication histories. We use three complementary designs. First, the native executions in Section 6 vary the time of reservation use, acknowledgement delivery and aggregate rewriting while retaining the original secret source. Second, two implementations of record readability are checked against each other and against independently stated expected results. Third, a reference corpus records the message keys derivable from actual component snapshots under fixed communication schedules. Table 4 relates each design to the question it answers.

Record and key configuration	Assignments
Journal-derived operation-wrapping key	512
Independent inner and outer wrapping keys	128
Rewrapped representations of one origin	512
Nested aggregate with derived keys	4,096
One actual key reused across wrappers	16
Empty record and unrelated source	64
Key-derivation cycle requiring a disclosed seed	32
Total	5,360

Table 5 Exhaustive acquisition, key-disclosure and direct-source assignments for the declared finite graphs. The general equality theorem is proved in Section 4.4; these checks validate its executable realizations on the stated graph inventory.

7.1 Finite-record semantics

The specialization in Section 4.4 has two executable realizations: iterative decryption of available record objects, and evaluation of normalized paths after computing key reachability. Their common inputs are a finite immutable record graph, canonical key identities, fixed public key-derivation edges, acquired records, directly disclosed keys, and directly disclosed source values. The evaluator checks equality of their exposed-source sets for every subset assignment in each declared graph.

Table 5 gives the complete finite inventory. The 5,360 assignments include direct-source subsets; they are separate from the 336 reference endpoint-snapshot cases reported above. Cyclic key derivation requires an initially disclosed key to seed reachability. Cyclic record containment is rejected. The source-preserving copy and rewrapping relations are inputs to these graphs, while the native executions in Section 6 provide their separately identified implementation evidence.

The history driver additionally evaluates all 360 legal permutations of a six-event acquisition, disclosure and deletion schedule, checking equality and knowledge monotonicity at all 2,160 prefixes. Ten directed histories exercise late key disclosure, capture before local deletion, retained pending targets, sibling copies, independent inner guards, and direct entropy disclosure. All records in these finite histories are initially available; record-creation events and operation-context validation remain in the surrounding provenance interface.

An independent review driver adds 17 cases with explicitly specified expected outputs. These cover seeded and unseeded key cycles, a self-loop, a reversed key-derivation chain, shared-child paths, direct child capture, partial keys from distinct paths, and local deletion of an embedded record's standalone copy. Expected outputs are checked independently of the agreement between the two algorithms. All 17 cases pass.

Five semantic controls exhibit the effect of discarding an adversary's copy on local deletion, fixing exposure at capture despite later key disclosure, resetting a source to the newest image identifier, dropping an inner-key guard, and treating a public

Component/profile	Mean bytes/send	Peak frame	Epochs A/B	Derived keys
SPQR, 32-byte chunks	32.150	37	24/23	172
SPQR, 64-byte chunks	58.971	69	46/45	88
Whole KEM, interval 1	819.658	1,225	683/683	4
Whole KEM, interval 32	44.047	1,223	32/32	66
Whole KEM, interval 64	25.023	1,223	16/16	130

Table 6 Component-level alternating workload; SPQR byte labels denote polynomial chunk width. Byte counts are the component wire contribution; epoch counts are each component's root/output observations. The constructions have different authentication and retention graphs. Outer application AEAD, transport, durable storage and energy are outside these measurements.

digest as record acquisition. Each produces a specified disagreement with the declared semantics. They are controls on interpretations of the record model.

Appendix A identifies the source, result files and independent expected-result checks. The reproduction entry point rejects execution modes that disable its assertions.

7.2 Reference snapshots and communication schedules

The pinned SPQR source is `f2589fef855c10f39d72634dab3d14654dd410bf`. Seven schedules each contain 2,048 sends: alternating, loss, reordering, duplication, 9:1 asymmetry, offline-then-exchange, and one-way traffic. Six capture points for each endpoint produce 84 passive snapshot cases. An authenticated whole-KEM control runs those schedules at local-send intervals 1, 32 and 64, producing 252 corresponding cases. Together these are 28 configuration–schedule combinations and 336 snapshots. An isolated 64-byte polynomial-chunk SPQR variant has a separate source identity; its alternating-workload result is explicitly labeled in Table 6. The 32-byte and 64-byte labels describe the polynomial chunk width.

7.3 Exposure and communication observations

The snapshot predictor receives one endpoint's serialized state and the full public transcript, including packets lost before peer delivery. A separate harness compares every derived key with the actual sender's key. Public deterministic seeds are test fixtures; the measurement establishes causal key derivation through the declared predictor interface. The output label `not_derived` denotes that predictor's result. It is separate from the security predicate of a computational experiment.

Table 6 reports the alternating-schedule observations. The snapshot is taken at A immediately after its first send, leaving 2,047 future sends. The pending KEM key accounts for 87 of the 172 derived keys in the pinned 32-byte SPQR case, demonstrating why the currently emitted chain epoch alone does not describe exposure.

The one-way schedule produces no fresh PQ epoch in these references and the predictor derives every post-snapshot key. The whole-KEM active-control experiment addresses a different property: one initial endpoint-state capture allows two attacker forks to replace subsequent exchanges while honest confirmed epochs increase. All 1,536 sender keys across three 512-send profiles match the attacker-derived keys. This supplies

a concrete distinction between fresh KEM execution and authenticated recovery when the ongoing authentication authority is itself derived from captured session state.

8 Consequences and further instantiation

A recovery condition attached to original operations.

The model distinguishes creation of an operation's random input, its first durable reservation, the acquisition of a retained representation, and the later availability of the matching decryption material. This distinction makes exact replay compatible with precise recovery statements: retries and aggregate writes preserve the original source, while fresh operations obtain new origins. The guarded characterization explains how records and keys combine to expose that source. The native executions demonstrate why both this conjunction and the original event history matter at the persistence and asynchronous-confirmation boundaries.

Joint root and message instantiation.

For the stated finite interface, Theorem 5.10 supplies a joint continuation through the actual confirmation, traffic and first-message schedule. Its private Expand-table normalization permits the traffic information to depend on root-derived confirmation tags, while producing the independent hidden input needed by the next reduction. The continuation retains actual confirmation and ACK verification, successor-root disclosure, and disclosure of the next chain seed. Exact retries share the encryption result from its first private computation.

The recovered-contribution entry uses the concrete root derivation of Equation (1),

$$p_e = \mathrm{HMAC}_{R_{e-1}}(S_e).$$

If R_{e-1} is disclosed, the remaining secret is in HMAC's message input, as captured by extractor and swap-PRF formulations [2, 15]. Appendix C.3 gives the shared-oracle Extract/Expand reduction for a uniform combined contribution. Composing it with the joint schedule and the specified one-use AEAD game bounds this profile's ideal-contribution score. The next instantiation step connects the actual accepted ContextBound operation to that uniform-contribution entry: the component challenge and complete-context hash hop must use the same original invocation, cache, and accepted-prefix history. The native correspondence term ρ_j and the selected candidate's actual projection residual $\eta^{\mathrm{nat}}_{\mathrm{proj}}$ retain their places in Proposition 5.8.

Matched protocol evaluation.

The reference corpus fixes reproducible communication opportunities for a matched Continuity comparison. Such a comparison records the same loss, reorder and duplication schedule, together with the construction's actual authentication authority and disclosure objects. For each case, the relevant observations are the original generation and encapsulation reservations, acquired record projection, known wrapping keys, public-wire availability, and reconstructed message set. Communication and persistence costs can then be measured over declared operation intervals. Holding a reservation

fixed while varying execution time, acknowledgement delivery or aggregate rewriting tests the provenance distinctions isolated by the present results.

Conclusion.

Persistent recovery is governed by original secret sources and their acquisition histories. The paper gives an exact record-exposure semantics, constructive simulations for private records and dependent immutable reports, a joint recovered-contribution-to-message continuation, and a mechanized multi-epoch recovery theorem with concrete separating executions. These results preserve the meaning of fresh randomness across durable retries, copies and asynchronous progress.

Data and code availability

The protocol source and recorded executions are identified by immutable repository revisions in Appendix A. The companion artifact contains the finite-record checker, expected-result cases, source maps, proof notes and complete manuscript sources. Each recorded result retains its source identity and execution receipt.

AI-assisted preparation

OpenAI ChatGPT was used for language editing, exploratory proof development, source-code inspection and LaTeX preparation. The explicit proofs, mechanized certificates and executable checks are identified separately so that their respective claims and verification records can be inspected.

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A Source and evidence map

The source snapshot for this manuscript's established protocol and experiment records is commit `228c944fc476e3e57d6c08063cacb19bca5b360e` of `billlza/q-periapt`. Each retained execution keeps its original source and tool identity. The local record model and accounting proofs in this manuscript are mathematical arguments; the following entries identify the independently recorded mechanization and native observations.

Result	Reproduction source
Joint symbolic replay	20261104-multi-epoch-replay. The original archive is `CAPTURE.zip`; the checked certificate member is `replay-01/01-baseline-replay.proof.spthy`.
Positive two-round trace	20261105-two-round-ground-trace.
Delayed-receipt ground trace	20261105-delayed-receipt-ground-trace.

Result	Reproduction source
Native reservation disclosure	RESERVATION_DISCLOSURE.md and `src/durable/messages/tests/reservation_disclosure.rs` in the identity candidate.
Aggregate-copy execution	20261105-aggregate-reservation-copy.
Root, traffic and phase mapping	RECOVERY_DEPENDENCIES_V1.md; native `rekey/response.rs` and `rekey/completion.rs`.
SPQR snapshot corpus	continuity-spqr-reference: `TRACE_CORPUS.json`, `COMPROMISE_CORPUS.json`, and `COMPROMISE_EXPERIMENT.md`.
Whole-KEM corpus	continuity-whole-kem-reference: `TRACE_CORPUS.json`, `README.md`, and `ACTIVE_COMPROMISE.md`.
Computational component arguments	COMPUTATIONAL_RECOVERY_V1.md and its linked KEM adapter, hidden-record, shared-HMAC, root-extraction and adaptive-target notes.

The jointly checked certificate has SHA-256

`dde5d06a071b1560da0d4c28c3f1fd81a3c39708f9fd8564ee391ddc362bcf7f`.

Its source-base receipt identifies `6084f3bfb09333f6b66291b38809ed512296a27b`. The fixed 4 November replay retained all 40 queries and its exact tool, resource and output receipts. Reading and checking this retained artifact while preparing the manuscript is distinct from executing another Tamarin run.

A.1 Mechanization and replay record

The theorem is the retained query `multi_epoch_fresh_reservations_recover_symbolic_message_secrecy`. Its certificate was replayed with all 24 reusable dependencies in the same Tamarin process. The independent 4 November replay used Tamarin 1.12.0 and Maude 3.5.1, one worker, a 12-GiB heap bound and the unchanged 900-second limit; it completed in 816.903 seconds. The output certificate was byte-identical to the retained input. Appendix A gives its digest and archive member, making the theorem's exact formula and proof available for inspection.

The complete inventory contains 40 queries: 27 verified and 13 incomplete. The latter comprise eleven multi-round existence queries and two historical all-trace queries outside the main theorem's dependency group. Two subsequent finite ground-trace constructions supply the concrete executions discussed below. Their independently checked formulas and original certificate inventory remain separately identifiable.

A.2 Guarded-normalization supplement

The complete manuscript extends the previously retained source at commit

`1c68aa666db22d36ee136be1ec712545a9e74f03`

under paper/continuity. The new material in this supplement is distributed locally with the present manuscript. Its filenames below are relative to this distribution; no remote publication location is assigned to them.

The explicit normalization theorem and its proof are in guarded-provenance.tex; the conjunctive-exposure proposition and proof are in acquisition-nonadditivity .tex. The executable supplement is in artifacts/guarded-provenance/:

- check_exposure.py evaluates both record semantics; results.json records the finite enumeration and history checks.
- review_exposure.py checks 17 independently stated expected results; review _results.json records their outcomes.
- review.md documents the theorem and implementation review; review_optim ized_guard.json records fail-closed optimized-mode rejection. The separately named review_optimized_run_before_guard.json retains the earlier execution that motivated the guard.
- native_source_map.json records source correspondence for the journal-to-wrapping-key derivation, durable image serialization, pending targets and the retained aggregate-copy test. It pins five source files at commit 228c944; this inspection records no new native execution.

From that artifact directory, the reproduction commands are

```
python3 check_exposure.py --output results.json
python3 review_exposure.py
```

These commands exercise the finite record model and its independent expected cases. The retained native and Tamarin evidence remains at the source checkpoints listed in the preceding map.

A.3 Constructive image/intent simulation

Section 5.1 integrates the complete finite-interface algorithm and proof from artifact s/record-rendering/opaque-renderer-v1.2.md. The corresponding mathematical review is in artifacts/record-rendering/independent-review.md. The retained source modules defining the commitment experiment are HIDDEN_RECORD_CO MMITMENTS_V1.md and MULTI_RECORD_COMMITMENTS_V1.md. Their actual domains and complete oracle budgets define the ϵ_{com} term; the new table construction supplies the image/intent implementation after that replacement.

The source correspondence of Section 5.4 is recorded in artifacts/native-int erface/native_interface_map.md and artifacts/native-interface/map.json. The machine-readable map binds the source ranges and file identities to the fixed protocol checkpoint. artifacts/native-interface/native_map_validation.json records the source checks. These are source observations associated with the declared interfaces; historical execution receipts retain their own dates and source identities.

A.4 Immutable report composition

Section 5.2 gives the finite-interface algorithm, reverse-order theorem and real-disclosure corollary. Its complete proof note is `artifacts/report-composition/multi_report_theorem.md`; the independent review is `artifacts/report-composition/multi_report_review.md`. The fixed-source CLEANUP_REPORT_MAC_V1.md records the fixed-source one-row interface. The paper states its operational game in Section 5.2 and its primitive instantiation in Appendix B. SHARED_ROOT_HMAC_V1.md specifies the compression-oracle functions and their query regimes. In the paper's notation, Φ_{256} bounds the two secret HMAC key-pad replacements, and E_{NMAC} bounds the ensuing row replacement with the stated length and query budgets. The composition theorem accepts the one-row distinguishing bound as a modular input and adds the complete ordinary-query work of every derived reduction.

A.5 Candidate-local computational connection

Section 5.3 incorporates the frozen proof fragment paired with `artifacts/candidate-local/candidate_local_composition.md` (version 0.2). The independent derivation and its artifact identities are recorded in `artifacts/candidate-local/independent_review.md` and `artifacts/candidate-local/independent_review.json`. They check the all-continuations stopping certificate, the exact native-score identity, the selected-candidate residual, and the local component comparisons. The score identity applies to bounded scores; the positive/negative-event conversion is specialized to the message experiment's three-valued score.

A.6 Joint schedule and live recovery profile

Section 5.5 and Appendix C give the joint schedule and its shared-oracle primitive derivations. The complete proof record is `artifacts/joint-continuation/joint_continuation.md`; its separate review and frozen source identities are retained in the same directory. The live recovery profile in `artifacts/live-profile/source_profile.md` maps the actual root, confirmation, traffic, message and ACK calls, together with the private application and persistent-record interfaces. The map and encoding inventory record the exact full information strings and source locations. Its 24 source files are byte-identical between the protocol pin `228c944f` and the inspected development revision `1c68aa66`; this identity check is separate from the retained native execution receipts.

A.7 ContextBound and original-operation inputs

The core `combine_policy_bound` and `combine_validated` functions in `crates/q-periapt-core/src/lib.rs` fix the length-delimited component and context encoding. The SDK replay constructor's `reserve`, `open` and `binding` operations in `crates/q-periapt-sdk/src/expert/replay.rs` retain the admitted operation and its original tape. The rekey response module's `kem_context` in `research/continuity-identity-candidate/src/durable/messages/rekey/response.rs` binds the complete retained offer into the application context. The artifact also retains an earlier Q-Periapt manuscript on ContextBound binding and implementation assurance. Those materials

document the reused cryptographic substrate; the original reservation, acquisition and persistent simulation results are developed in the present paper.

Relation to earlier project material.

Earlier project work on SkyBridge Compass studies policy-governed authenticated establishment, complete-transcript binding, and a control-only recovery phase before application traffic.[1] The connection is the separation of operation admission and release from the secrecy of the inputs used by that operation. Here we track original randomness through persistence and acquisition, and construct the associated record and report simulations. The artifact's `notes/skybridge-scope.md` records the exact source versions and the allocation of material between these research questions.

B Primitive inputs to the record and report bounds

The composition theorems use operational distinguishing bounds: a record commitment comparison and a single immutable report row. This appendix fixes the concrete parameter conventions. The public hash and compression oracles are idealized in the stated games; the source construction uses the specified SHA3 and HMAC-SHA256 algorithms.

B.1 Hidden-input random-oracle replacement

For a hidden-input entropy parameter n, define

$$\Phi_n(d, Q) = \min\left\{1, 4\sqrt{2d^2Q\,2^{-n}} + 4\sqrt{2Q^2d\,2^{-n}}\right\}.$$

Don, Fehr and Huang's Theorem 2 [11] supplies this bound when private oracle inputs have point probability at most 2^{-n} and distinct effective private queries encode distinct inputs for every secret value. Exact repeats are cached before that interface is queried. The two budgets count private classical queries and all ordinary public-oracle queries; public queries may be quantum.

For image i, let K_i^{hdr} count original intent prefixes and let $d_i \le 1 + 5K_i^{\mathrm{hdr}}$ count its distinct private digest inputs. There is one image-head domain and five intent-digest domains. With an independent 192-bit seal nonce, the fixed-input encoding and per-nonce confinement conditions give the retained record bound

$$\epsilon_{\mathrm{com}} \le \min\left\{1, \sum_{i=1}^{M} \Phi_{192}(d_i, Q_i)\right\}.$$

Here $d_i = 0$ for known, atom-free rows. The budget Q_i includes the entire derived distinguisher, all other real rows and the final test. The nonce remains private before the declared acquisition stop. Local image encryption and intent authentication introduce no additional shared SHA3 calls. This comparison is made after private-wire equality

[1] Unpublished project manuscript, source revision `d49d2a9`.

has been replaced by structural equality; its separate collision term is the one in Corollary 5.3. The exact encoding domains and fixed-source arguments are identified in Appendix A.3.

B.2 One immutable HMAC row

For the row experiment in Section 5.2, let

$$h : \{0,1\}^{512} \times \{0,1\}^{256} \longrightarrow \{0,1\}^{256}$$

be the public compression oracle, with arguments in message-block, chaining-state order. A 32-byte key K determines the two pad blocks

$$I(K) = (K\|0^{32}) \oplus \mathtt{36}^{64}, \qquad O(K) = (K\|0^{32}) \oplus \mathtt{5c}^{64}.$$

Zero exponents in these encodings count bytes. The ranges are disjoint and each encoding is injective in K. Cache the two evaluations at the fixed SHA-256 initial state. All oracle-dependent preparation is executed and counted; no oracle-correlated advice is supplied for free.

Let L be the maximum message length in bytes and put

$$L \le 2^{61} - 65, \qquad \ell = 1 + \left\lceil \frac{L+9}{64} \right\rceil .$$

The SHA-256 padding encodes $8(64 + |m|)$ bits, retaining the already consumed 64-byte key-pad block. Its suffix is determined by $|m|$ and it is injective within the stated length domain. The parameter ℓ bounds the remaining row compression work after the two key-pad calls.

Write $E_{\mathrm{NMAC}}(T, q, \ell)$ for a valid NMAC distinguishing bound with two independent 256-bit inner/outer keys, the prescribed injective padding, at most T keyed calls and q public compression calls. Hosoyamada and Iwata's Proposition 6 [14] gives the asymptotic form

$$E_{\mathrm{NMAC}}(T, q, \ell) = O\left(\sqrt{\frac{(q+T)^3 \ell^5}{2^{256}}}\right)$$

under its stated padding and query conditions. The implicit constant remains part of that published bound. Our message-length convention retains HMAC's consumed-block offset.

The two key-pad replacements use $\Phi_{256}(2, q + \ell T)$. The subsequent NMAC replacement gives an admissible row bound

$$\epsilon_{\mathrm{row}}(T, q, L) = \min\{1, \Phi_{256}(2, q + \ell T) + E_{\mathrm{NMAC}}(T, q, \ell)\} .$$

This is a restricted single-transcript application of the primitive results. The ideal NMAC function is queried on one original transcript only, so its first reply is a uniform tag and exact repeats retrieve the same value. Report nonzero-ID checks act on that

reply in both games. All other real HMAC work remains in q; the multi-report theorem supplies the dependency order and its actual disclosure residual.

C Shared-oracle primitives for the joint continuation

This appendix supplies the primitive reductions used in Section 5.5. They use the same idealized SHA-256 compression oracle throughout. Each experiment begins with independent oracle, hidden uniform 256-bit input and auxiliary state. All oracle-dependent preparation is executed within the counted algorithm. The designated hidden value has only the stated interfaces, and neither it nor the private random tables are returned to the terminal test. Ordinary public queries may be quantum; private function queries are classical and exact repeats are cached.

We use Φ_{256} and E_{NMAC} from Appendix B, with

$$\ell(L) = 1 + \left\lceil \frac{L+9}{64} \right\rceil, \qquad L \le 2^{61} - 65.$$

The HMAC padding includes the consumed 64-byte key pad: its encoded length is $8(64 + |m|)$. Equal padded strings have equal encoded lengths and hence equal message prefixes within this domain. Thus the required padding is injective, and $\ell(L)$ bounds the ordinary row compression work after its two key-pad evaluations.

C.1 A shared HMAC row and column

For a 32-byte value x, define the 64-byte blocks

$$\begin{aligned} I(x) &= (x \| 0^{32}) \oplus \mathtt{36}^{64}, \\ O(x) &= (x \| 0^{32}) \oplus \mathtt{5c}^{64}, \\ B(x) &= x \| P, \qquad P = \mathtt{80} \| 0^{23} \| \,\mathrm{BE64}(768). \end{aligned}$$

The final 32 bytes of $I(x), O(x), B(x)$ are respectively $\mathtt{36}^{32}, \mathtt{5c}^{32}, P$. These three ranges are disjoint, and each map is injective. Write

$$\alpha(a) = h(I(a), \mathrm{IV}), \qquad \beta(a) = h(O(a), \mathrm{IV}), \qquad \Gamma(a) = (\alpha(a), \beta(a)).$$

The actual 32-byte-message column is

$$F(a, R) = h\big(B(h(B(R), \alpha(a))), \beta(a)\big). \tag{15}$$

The actual row is the NMAC-form expression

$$F(R, m) = \mathrm{Iter}_h\big(\beta(R), \mathrm{pad}_{64}\big(\mathrm{Iter}_h(\alpha(R), \mathrm{pad}_{64}(m))\big)\big),$$

where pad_{64} retains the consumed-key-pad length offset.

For completeness, $E_{F_1}(C, q)$ denotes a valid distinguishing bound for

$$F_1(u, v) = \widetilde{h}(v, f(u))$$

versus a private random function of the complete pair (u, v). Here f is an independent private random function from 256 to 256 bits; $\widetilde{h}$ is a public random function on 512-bit v and 256-bit final input; there are C classical F_1 calls and q ordinary quantum queries. Hosoyamada and Iwata's Proposition 4 [14] gives

$$E_{F_1}(C, q) = O\left(\sqrt{\frac{q_*^3}{2^{256}}}\right), \qquad q_* = \max\{C, q\} = o(2^{256/3}),$$

with the published constants and regime retained.

Lemma C.1 (Independent-table shared-root bound) *An admissible operational bound is*

$$\begin{aligned}\varepsilon_{\mathrm{Hjoint}}(C, T, q, L) \le \min\{1,\ & \Phi_{256}(C + 2, q + 3C + \ell(L)T) \\ & + E_{\mathrm{NMAC}}(T, q + 3C, \ell(L)) \\ & + E_{F_1}(C, q + 2C)\}.\end{aligned} \tag{16}$$

The ideal row and column are independent private functions $U(m)$ and $V(\Gamma(a))$, independent of the public oracle.

Proof First isolate every compression call with a raw occurrence of R:

$$\mathcal{F}_R(t, s) = h(\tau_t(R), s), \qquad \tau_{\mathrm{in}} = I, \quad \tau_{\mathrm{out}} = O, \quad \tau_{\mathrm{msg}} = B.$$

For a fixed (t, s), each encoded point has probability at most 2^{-256}. Different effective pairs encode different points for every R: different tags have disjoint block ranges, and equal tags retain the chaining state. Cache by the complete pair (t, s), not just by the salt. This matters when two salts have the same α value.

There are at most $C+2$ distinct private calls: the two root key-pad evaluations and at most C message-block evaluations at $\alpha(a)$. The remaining ordinary work is at most $q + 3C + \ell(L)T$, including each column's two public pad states and outer call, and each row's continuation. Don, Fehr and Huang's Theorem 2 [11] therefore replaces $\mathcal{F}_R$ by a private random function at the first cost in (16).

Its restrictions

$$A = \mathcal{F}(\mathrm{in}, \mathrm{IV}), \quad D = \mathcal{F}(\mathrm{out}, \mathrm{IV}), \quad f(s) = \mathcal{F}(\mathrm{msg}, s)$$

are independent of one another and h. The row is now NMAC with independent inner and outer keys A, D. The simultaneous column is $h(B(f(\alpha(a))), \beta(a))$. To replace the row, the NMAC reduction samples its own f table and answers a column with three ordinary compression calls. Proposition 6 of Hosoyamada and Iwata [14], with the prescribed offset padding, gives the second term and a private row function U independent of f, h.

For the column, reindex the *same* oracle by

$$\widetilde{h}((t, s), y) = h(y \| t, s), \qquad |t| = |s| = |y| = 256.$$

This is a fixed bijection on all 768 input bits. It is implemented coherently by wire permutation and one oracle call, not by introducing an independent hash. The remaining column is exactly

$$\widetilde{h}\big((P,\beta(a)),f(\alpha(a))\big)=F_1\big(\alpha(a),(P,\beta(a))\big).$$

Computing its public states costs two ordinary calls per column. The independent row table U needs no compression queries. The F_1 bound therefore applies with C private calls and at most $q+2C$ ordinary calls. Its ideal reply is indexed by $(\alpha(a),P,\beta(a))$, equivalently by $\Gamma(a)$ since P is fixed. Equal Γ values must share a reply; their equality is never assumed to imply equal salts. Summing the three replacements proves the result. □

The lemma compares against the independent-table ideal. In particular it does not assert pointwise real equality at an exceptional cross-interface pair where both classical arguments equal the hidden root. Such guesses are admitted by the real experiment and covered by the three distinguishing steps. A different ideal that explicitly repairs this cross-pair needs its own comparison; it is not used here.

C.2 The multiple-message row and a single Extract column

For $\varepsilon_{\mathrm{Hrow}}(T,q,L)$, the adversary has an adaptive classical row oracle. It can choose later messages using earlier replies and its quantum computation with the common public oracle. Equal messages retrieve equal values. There is no requirement that the full transcript be fixed before the first reply, and there is no raw-key disclosure channel.

Lemma C.2 (Row and single-column instances) *The following bounds instantiate the operational games:*

$$\varepsilon_{\mathrm{Hrow}}(T,q,L)\le\min\{1,\ \Phi_{256}(2,q+\ell(L)T)+E_{\mathrm{NMAC}}(T,q,\ell(L))\},\tag{17}$$

$$\varepsilon_{\mathrm{Hcol}}(q)\le\min\{1,\ \Phi_{256}(3,q+3)+E_{F_1}(1,q+2)\}.\tag{18}$$

Proof For the row, only $I(K)$ and $O(K)$ contain the hidden key. The same disjoint-encoding replacement uses two private calls and at most $q+\ell(L)T$ ordinary calls. The resulting independent-key NMAC receives all T adaptive messages, and its replacement is precisely the private row function. No conversion to a one-transcript game is made.

For one distinct zero-salt column, omit the NMAC step from Lemma C.1, put $C=1$, and use its conservative $C+2=3$ private-call budget. The only visited $\Gamma(0)$ entry gives one uniform reply independent of h and of the executed prefix. The extra two private-call slots need not actually be used; retaining this upper bound agrees with the existing shared-root accounting. □

C.3 Root initialization at the actual entry boundary

Let S be an independent uniform 32-byte contribution. Executed preparation may choose a known 32-byte old root r and the response core using the public oracle, but has no other raw use of S. Consider one original Extract/Expand,

$$P=F(r,S),\qquad R=F(P,I_R\|\mathtt{01}),$$

with the actual 97-byte information string from Section 5.5. Repeats reuse this result. The value P has only this Expand use; S, P have no other output channel. The entire later protocol is generated from R.

First apply the single hidden-message column argument to salt r. Its ideal value is $V(\Gamma(r))$. Even when r depends on the public oracle prefix, this first table value is fresh independent uniform: conditional on each oracle and prepared state, its index is fixed and has not been visited. Equivalently presample a fresh PRK and withhold it. This does not condition on a secret-dependent public transcript.

Embed a hidden row key at that PRK and answer the one 98-byte Expand message. Its ideal reply is a fresh uniform root. It can again be presampled and withheld until creation, independent of the common oracle and all earlier state. All later confirmations, signatures, traffic and successor operations are recomputed from this root. The proof therefore supplies the initialization for Theorem 5.10 to an entire interactive continuation, not to a marginal root distribution.

If B bounds all ordinary compression work other than this derivation and the displayed public salt-state computations, conservative budgets are $q_{\mathrm{ext}} \le B + 5$ and $q_{\mathrm{exp}} \le B + 2$. The former includes the five actual root-Expand compression calls; the latter includes the two public states in $\Gamma(r)$. Since $\ell(98) = 3$, the resulting bit error is

$$\begin{aligned}\Delta_{\mathrm{init}} \le \min\{1,\ & \Phi_{256}(3, B+8) + E_{F_1}(1, B+7) \\ & + \Phi_{256}(2, B+5) + E_{\mathrm{NMAC}}(1, B+2, 3)\}.\end{aligned} \tag{19}$$

All terms count the complete continuation in their respective reductions. The published asymptotic constants are not converted into a numerical security level.

The downstream length budgets are equally explicit: confirmation information lengths $62, 58, 60$ give row messages $63, 59, 61$; the 155-byte traffic information gives $156, 188, 188, 188$; and the 56-byte first-step information gives $57, 89$. Thus the shared root row has $\ell(32) = 2$, the traffic rows have maximum $\ell(188) = 5$, and the selected chain row has $\ell(89) = 3$. These counts include the previous HKDF output block where it is part of the actual HMAC message.

D Accounting through one common virtual experiment

This appendix records the alternative accounting route when one common real-component virtual experiment and a global native-to-virtual comparison have been established. In that setting the shared comparison cost is charged once; uniform resource bounds follow by direct substitution. Section 5.3 gives the candidate-local route used by the record and report interfaces.

D.1 A selected-target experiment

Let V be the virtual original-record experiment with the real KEM component. The adversary chooses its target after its observed interaction. Eligibility E retains the selected recovery conditions: both original target reservations follow Cut, the actual target signing keys and target-epoch entropy remain unexposed, and no selected-session

record saved after Cut is acquired. The experiment follows allowed asynchronous control and application transitions, including sending after the local FinalReady transition.

Let Attr mean that the selected target has a unique usable pair of original slots, and let $I \in J$ denote that pair. The mapping and every event used by the reduction must be computable from the adapter's known semantic state and observations. Let

$$\eta \;\geq\; \Pr_{V}[E \wedge \neg \mathsf{Attr}].$$

For $j \in J$, define V_j as the full experiment with the adapter's random component override at pair j. Its transcript, eligibility and target choice are recomputed in that world. The remaining protocol and oracle calls use the same common interfaces.

The application test bit is sampled by the surrounding simulator. Write $S = +1$ for a correct guess and $S = -1$ otherwise; invalid tests contribute zero through E. Every $E = 0$ execution is completed by an independent fair guess. Thus the usual message-guessing advantage is $\frac{1}{2}|\mathbb{E}[ES]|$. The simulator can evaluate this score without observing the hidden bit of its component KEM challenger.

D.2 Centered-score composition

For each j, put $E_j = E \wedge \mathsf{Attr} \wedge (I = j)$. Suppose the fixed-pair component adapter gives, for $\sigma \in \{+1, -1\}$,

$$\left| \Pr_{V}[E_j \wedge S = \sigma] - \Pr_{V_j}[E_j \wedge S = \sigma] \right| \leq a_j^{\sigma} + \delta_j, \tag{20}$$

where a_j^{σ} is the advantage of the actual derived component distinguisher and δ_j is the corresponding correctness term. The distinguisher includes target selection and all relevant oracle queries. Define

$$b_j = \mathbb{E}_{V_j}[E_j S].$$

Finally, let ϵ_{rec} bound the binary distinguishing gap between the native and virtual experiments for their efficiently evaluable positive-score and negative-score events. This is the concrete record-interface connection, including public outputs and accepted operation histories.

Proposition D.1 (Adaptive origin-pair accounting) *Under these stated interface and fixed-pair bounds, the native application's guessing advantage satisfies*

$$\mathrm{Adv}_{\mathrm{msg}} \leq \min \left\{ \frac{1}{2},\; \epsilon_{\mathrm{rec}} + \frac{1}{2} \left(\eta + \sum_{j \in J} |b_j| + \sum_{j \in J} (a_j^{+} + a_j^{-} + 2\delta_j) \right) \right\}. \tag{21}$$

Proof In the single experiment V, the attributed events E_j are disjoint. Decomposing its score yields

$$\mathbb{E}_V[ES] = \mathbb{E}_V[ES\,\mathbf{1}_{\neg \mathsf{Attr}}] + \sum_{j \in J} \mathbb{E}_V[E_j S].$$

The first term has magnitude at most η. Each attributed expectation is the positive-event probability minus the negative-event probability. Applying (20) to those two events and then the triangle inequality bounds its distance from b_j by $a_j^+ + a_j^- + 2\delta_j$. Summing gives the virtual score bound. The two native/virtual binary-event comparisons contribute at most $2\epsilon_{\mathrm{rec}}$ to the score difference. Dividing by two and using the maximum bit-guessing advantage proves (21). □

The selector may depend adaptively on its entire observed transcript. The absolute values in $\sum_j |b_j|$ retain each override world's own target selection. In particular, the override probabilities are not normalized as though they belonged to one common ideal distribution. The score decomposition also retains failed and ineligible executions in the experiment, avoiding an uncontrolled division by the probability of eligibility.